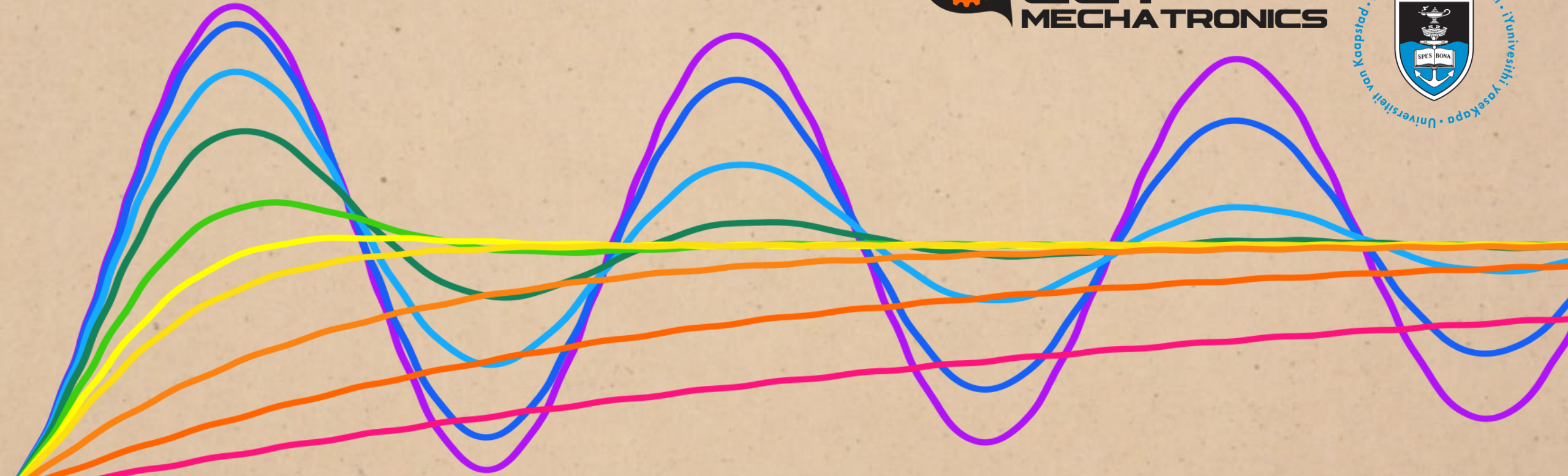




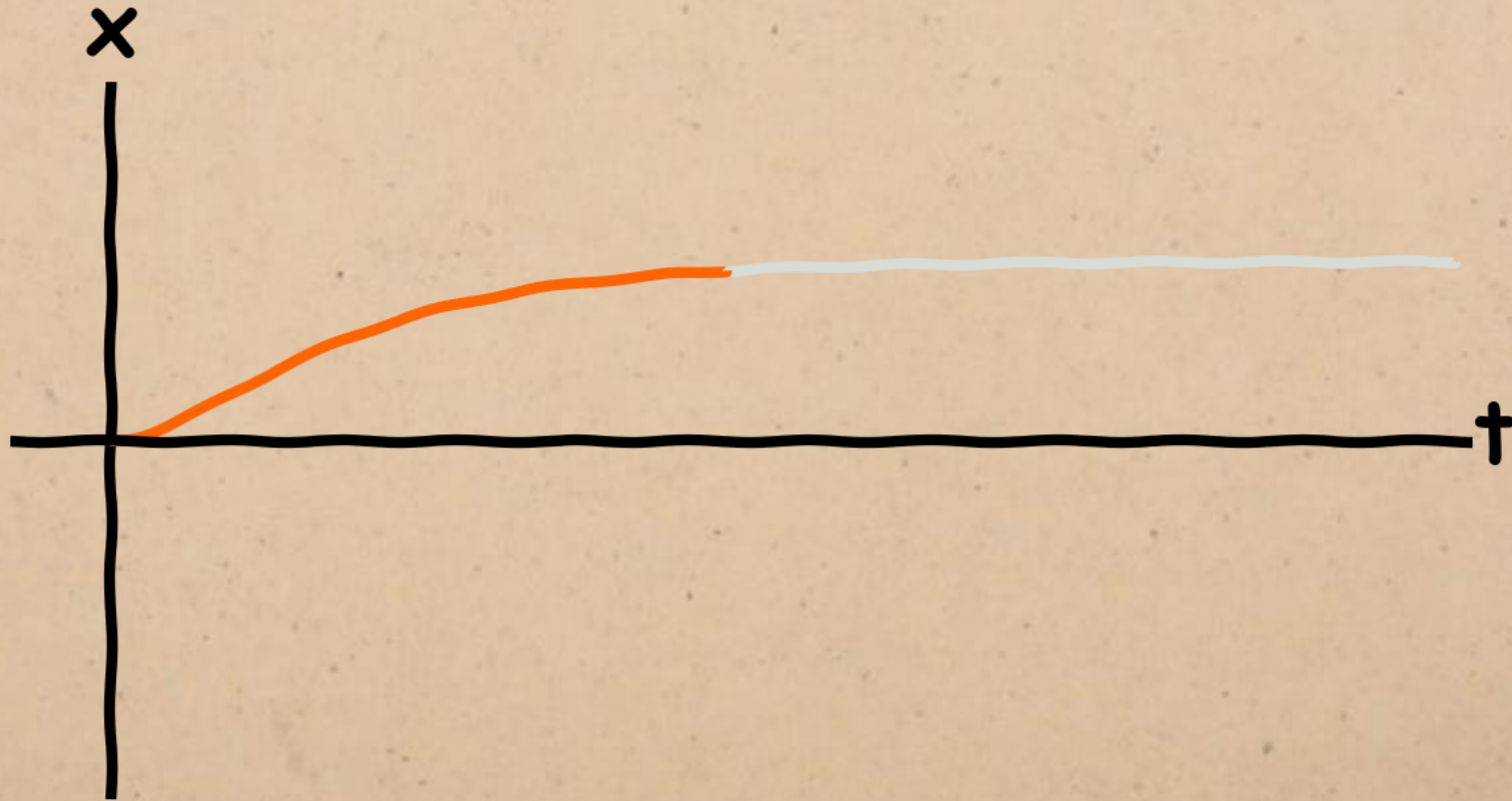
CHAPTER 2: SYSTEM DYNAMICS

TRANSIENT RESPONSE

PART I

LAST TIME:

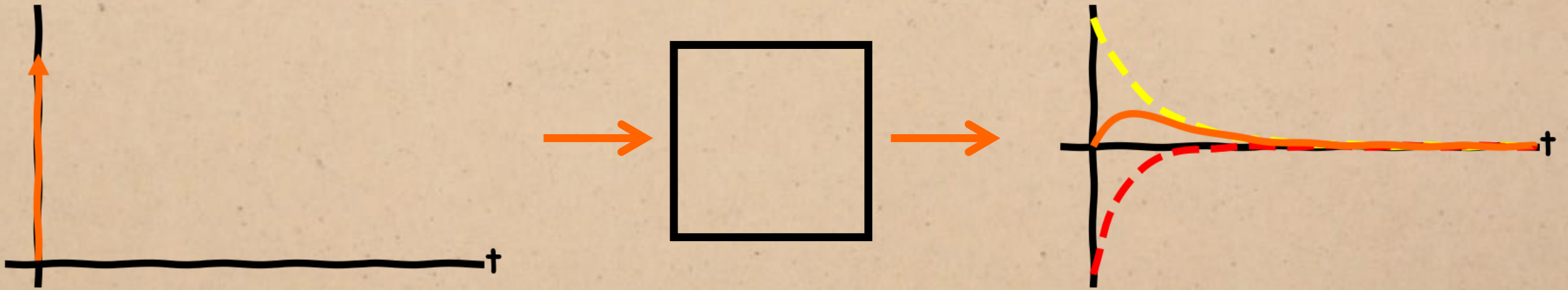
transient response = part of the response that decays away over time



LAST TIME:

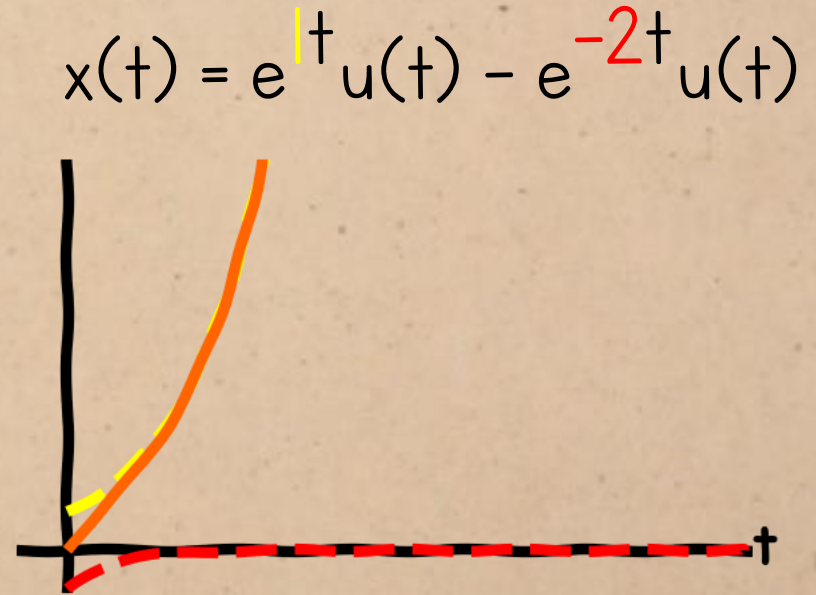
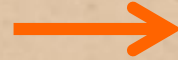
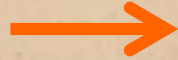
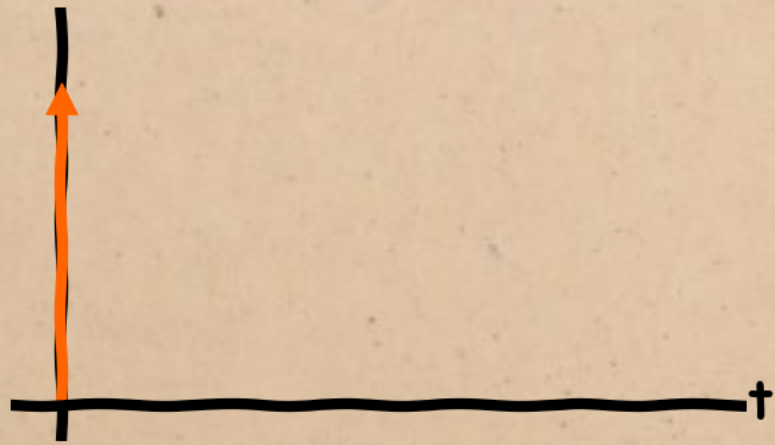
The transient response dies away because all the exponential modes are decaying.

$$x(t) = e^{-1t}u(t) - e^{-2t}u(t)$$



LAST TIME:

If any of the modes are growing, the transient response will also grow to infinity.



LAST TIME:

characteristic equation = denominator of
transfer function set equal to zero

$$x(s) = \frac{1}{(s + 6)(s + 4)} F(s)$$

$$(s - -4)(s - -6) = 0$$

LAST TIME:

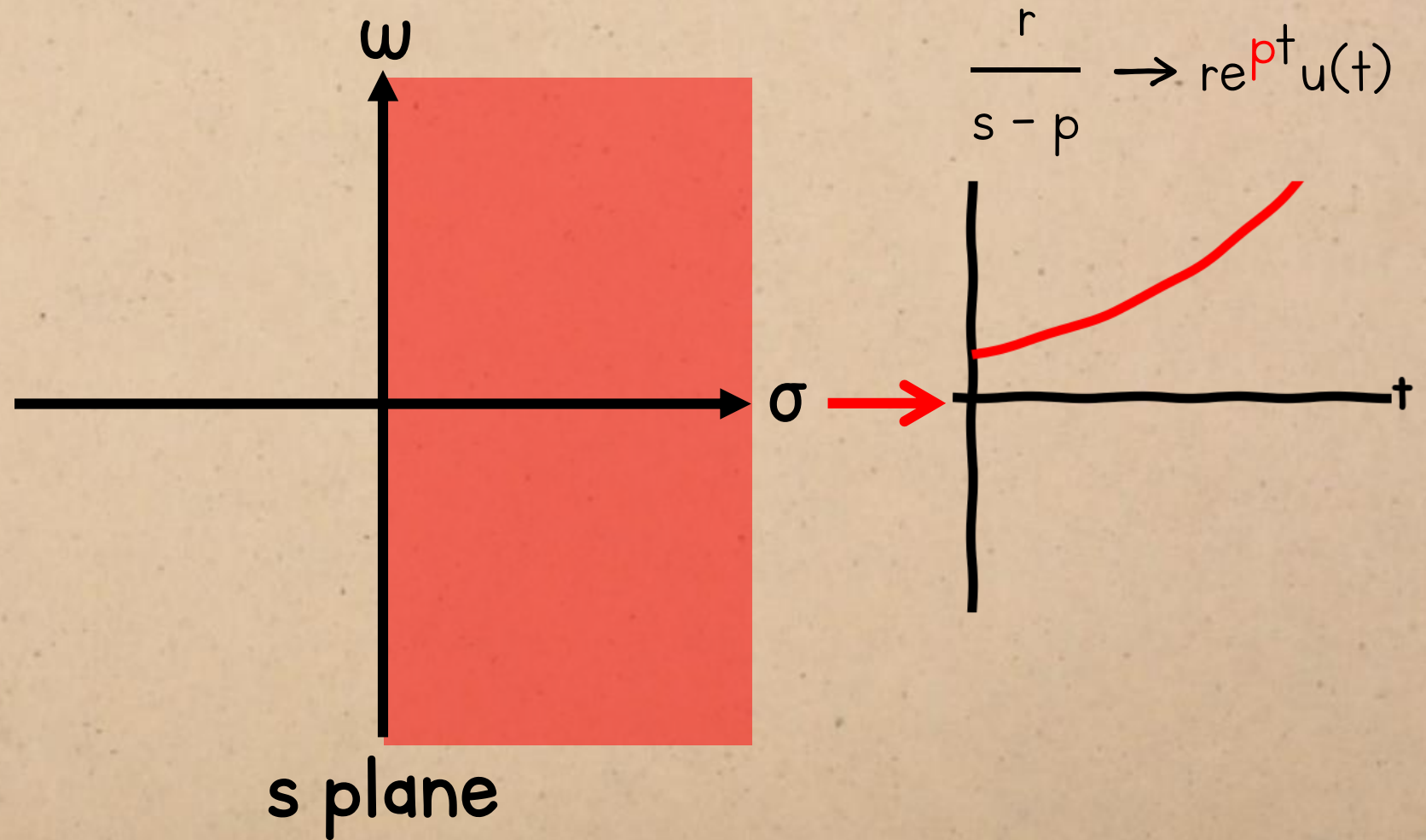
poles = roots of the characteristic equation

$$x(s) = \frac{1}{(s + 6)(s + 4)} F(s)$$

$$(s - -4)(s - -6) = 0$$

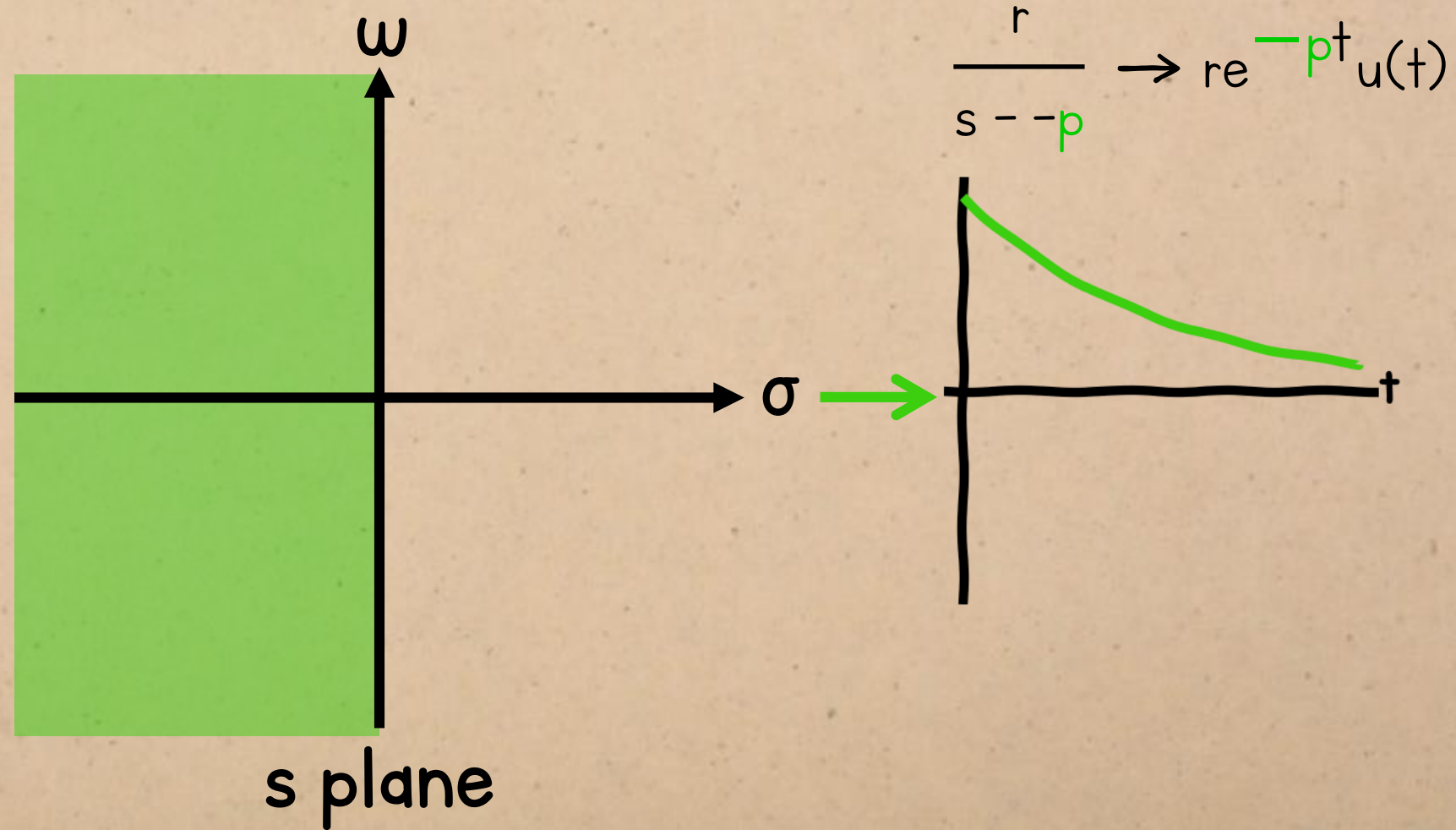
LAST TIME:

Poles in the right-half plane (negative poles) are **unstable**.

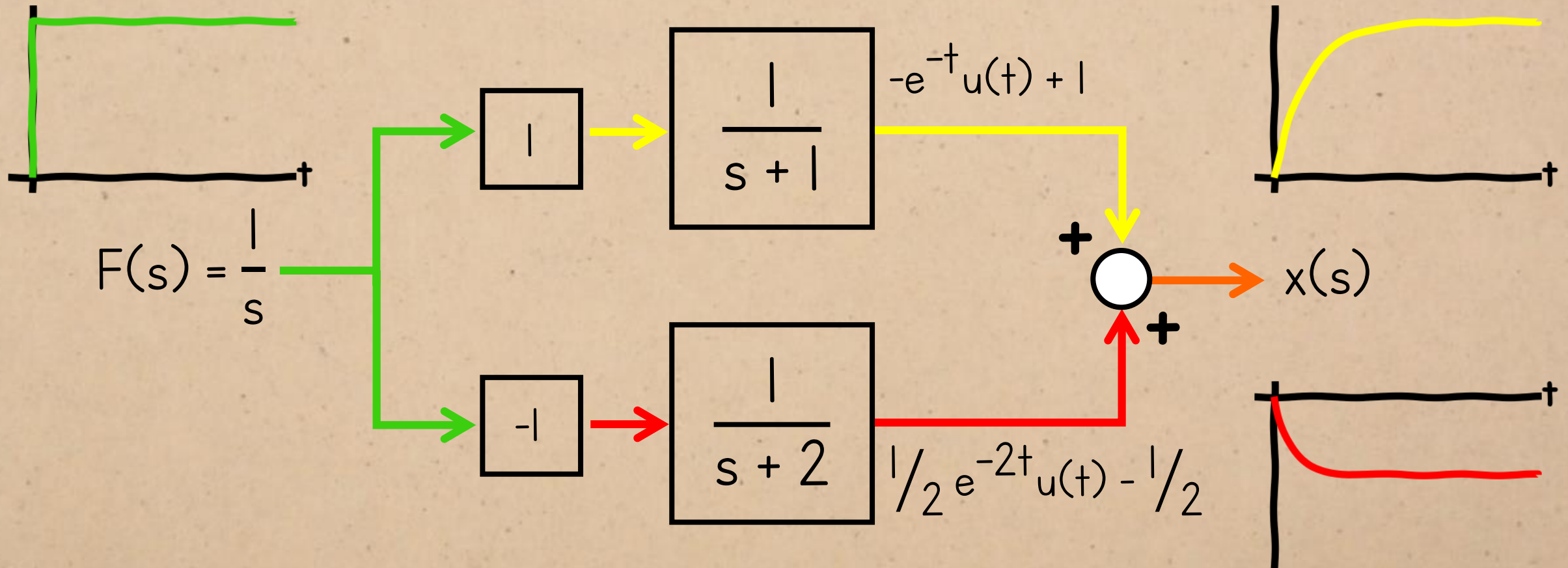


LAST TIME:

Poles in the **left-half plane** (positive poles) are **stable**.



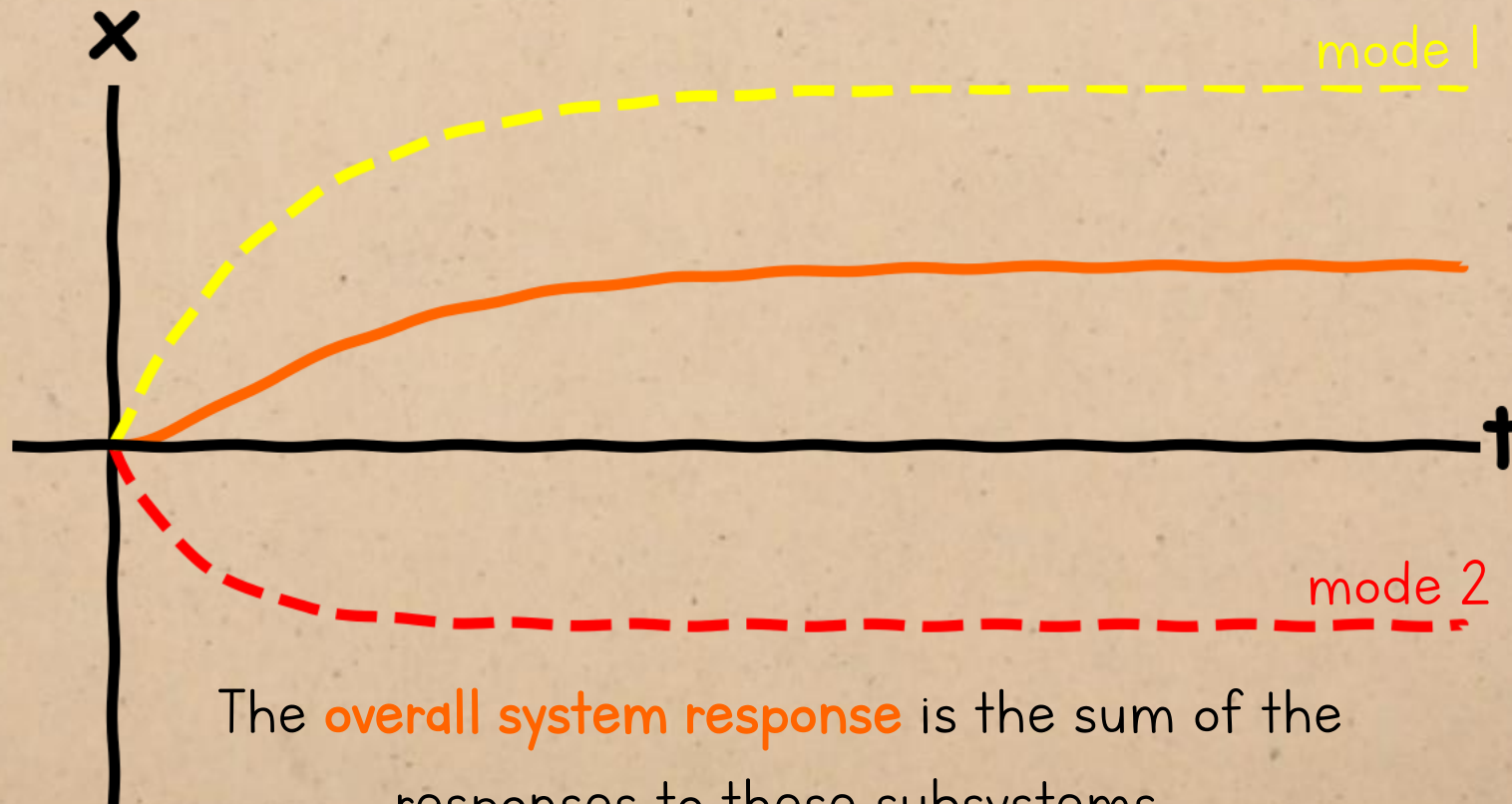
LAST TIME:



We can decompose a system into first-order components.

LAST TIME:

$$x(t) = e^{-t}u(t) - \frac{1}{2}e^{-2t}u(t) + \frac{1}{2}$$



The **overall system response** is the sum of the responses to these subsystems.

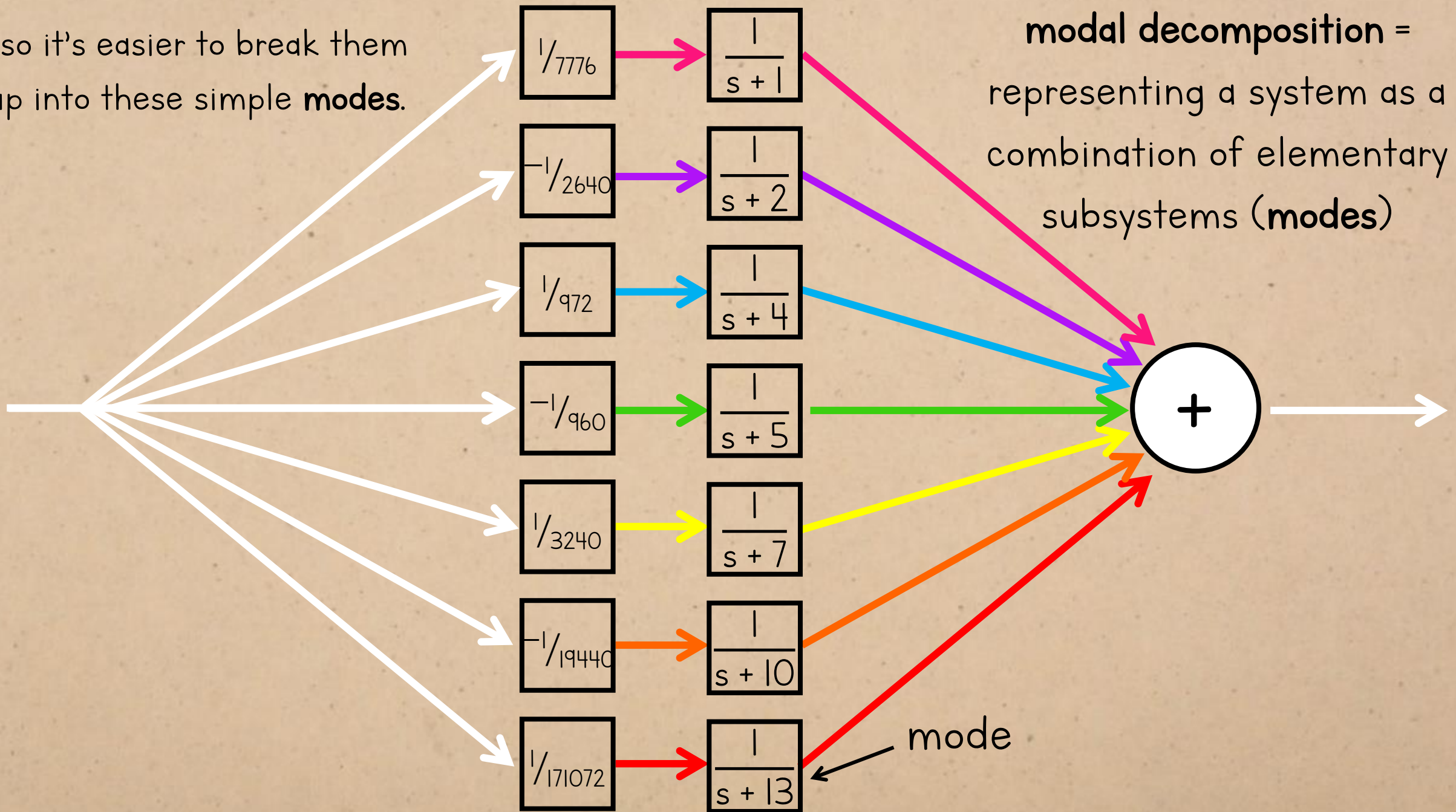
LAST TIME:

High-order systems are difficult to work with...

|

$$s^7 - 42s^6 + 700s^5 - 5950s^4 + 27559s^3 - 68488s^2 + 82620s - 36400$$

...so it's easier to break them up into these simple **modes**.



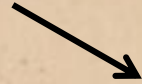
FIRST-ORDER SYSTEM

$$\frac{A}{\tau s + 1}$$

This lecture will look at the different modes we can use to build systems, starting with the simplest one: the first-order system.

FIRST-ORDER SYSTEM

gain



A



$\tau s + 1$

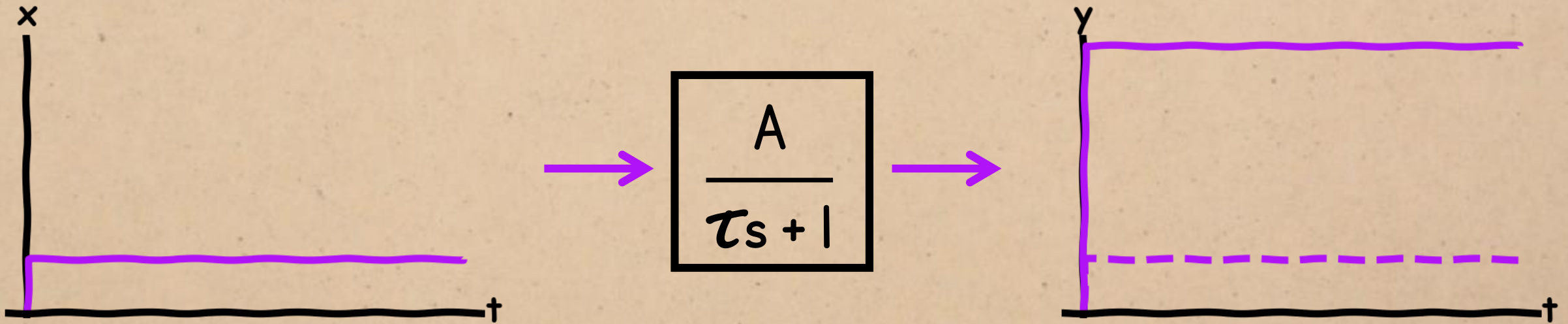
FIRST-ORDER SYSTEM

$$\lim_{s \rightarrow 0} \frac{A}{\tau s + 1} = A$$

We know A is the gain, because it's the value we get if we let s tend to zero.

FIRST-ORDER SYSTEM

$$\text{gain} = \frac{\text{height of output}}{\text{height of input}}$$



If we have an unknown first-order system, we can find the gain from its response to a unit step.

FIRST-ORDER SYSTEM

A

$\tau s + 1$



time constant

FIRST-ORDER SYSTEM

$$\frac{r}{s - -1/\tau}$$

residual $r = \frac{A}{\tau}$

Note: In the original image, the 'r' in the numerator is purple and the '-1/τ' in the denominator is teal.

We can rearrange the general expression for the first-order system into this form, which matches the Laplace transform table.

FIRST-ORDER SYSTEM

$$re^{-t/\tau}u(t)$$

...and then take the inverse Laplace transform, which gives the general, time-domain form of the first-order impulse response.

FIRST-ORDER SYSTEM

$$re^{-t/\tau}u(t)$$

The time constant determines the **rate of decay** of the time-domain exponential.

FIRST-ORDER SYSTEM

step response:

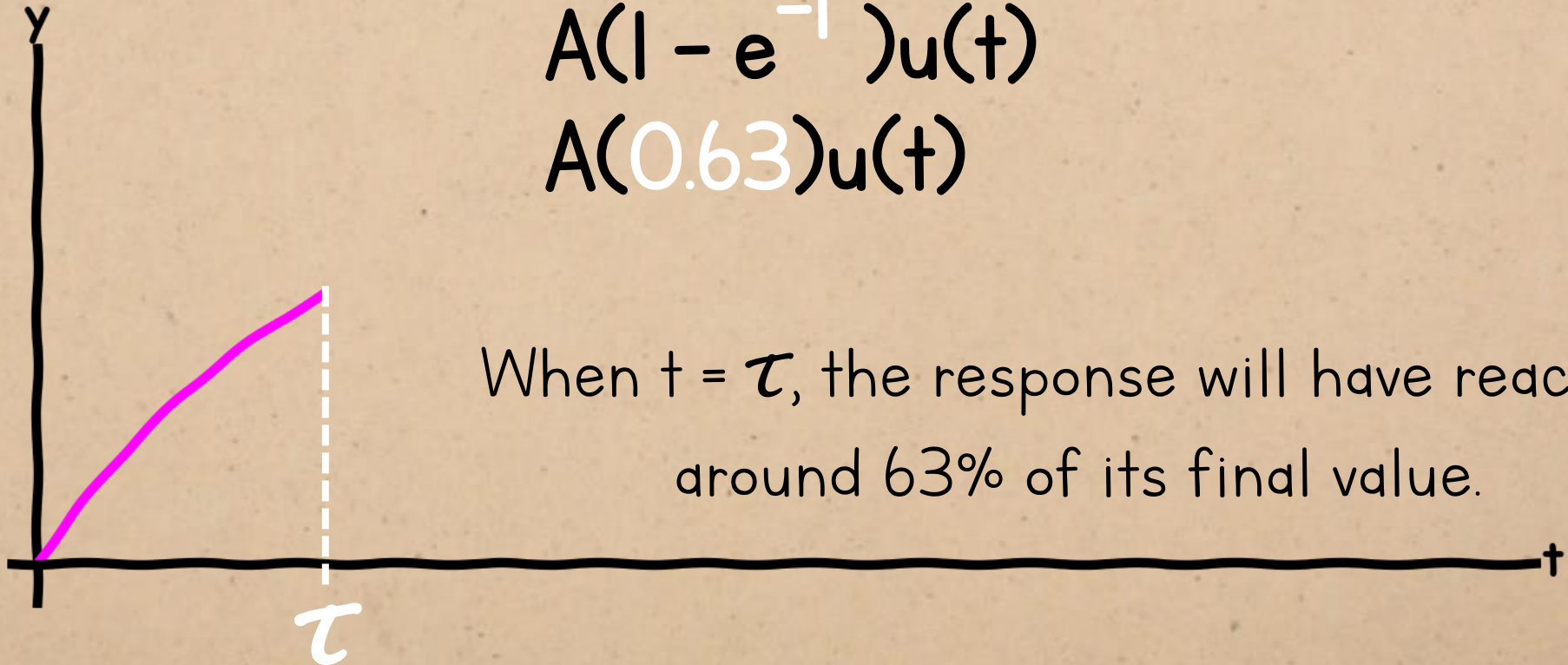
$$A(1 - e^{-t/\tau})u(t)$$

FIRST-ORDER SYSTEM

$$A(1 - e^{-t/\tau})u(t)$$

$$A(1 - e^{-1})u(t)$$

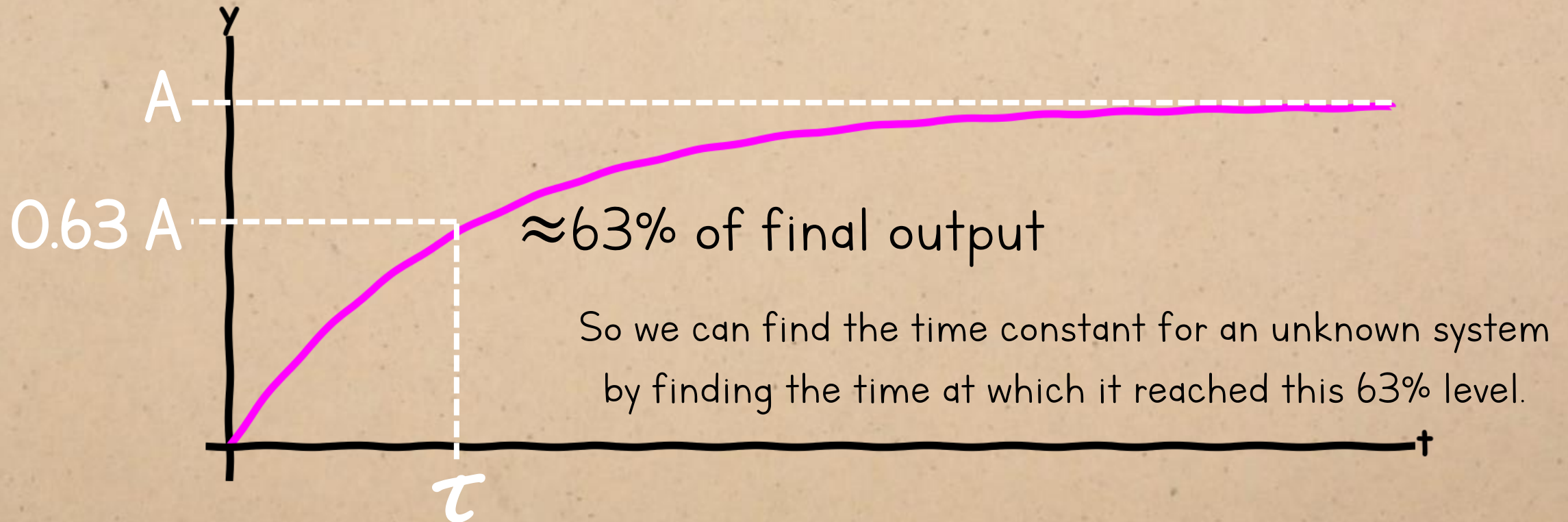
$$A(0.63)u(t)$$



When $t = \tau$, the response will have reached around 63% of its final value.

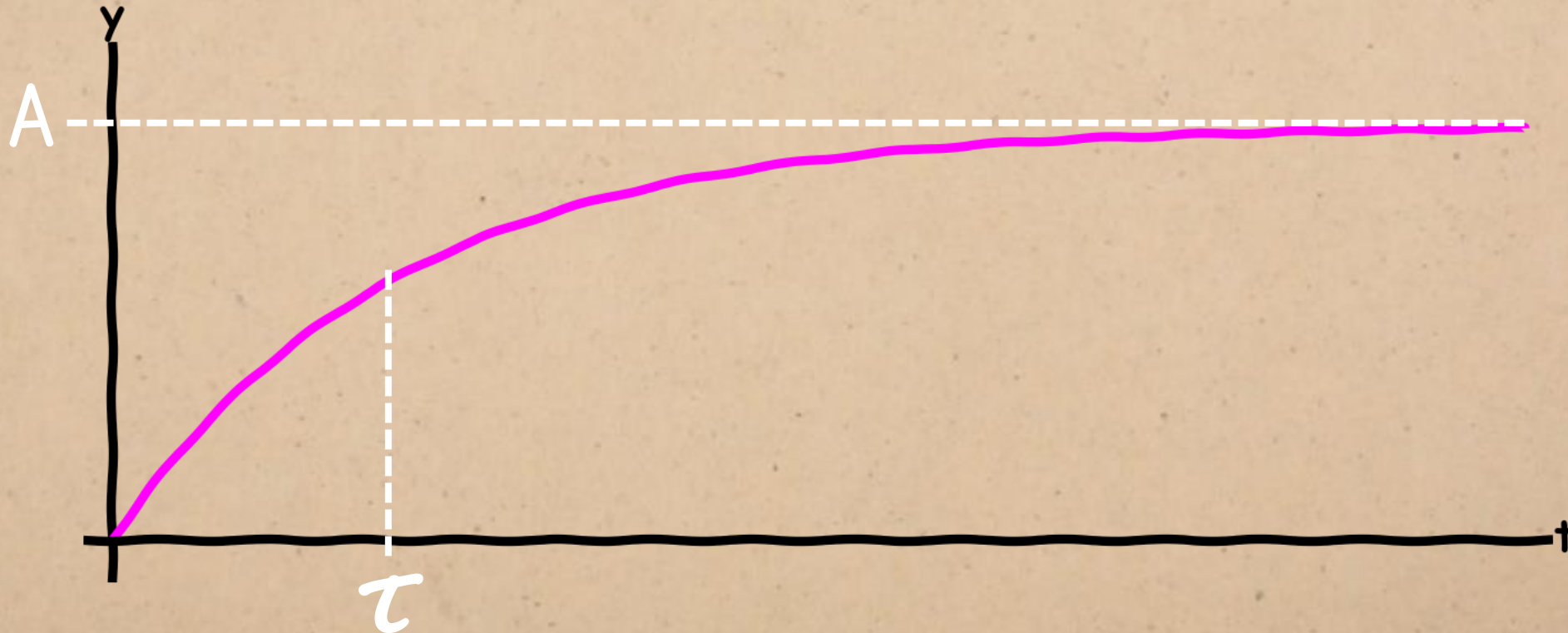
FIRST-ORDER SYSTEM

$$A(0.63)u(t)$$



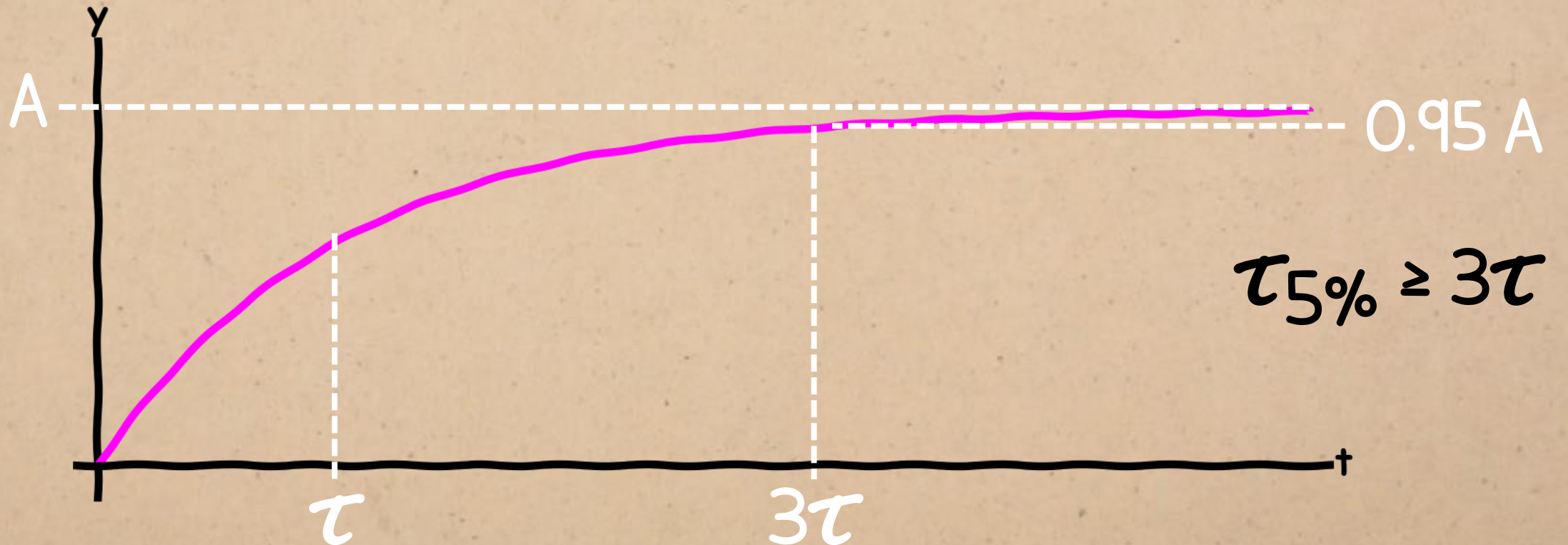
FIRST-ORDER SYSTEM

settling time = time it takes for response to get within some % of its final value.



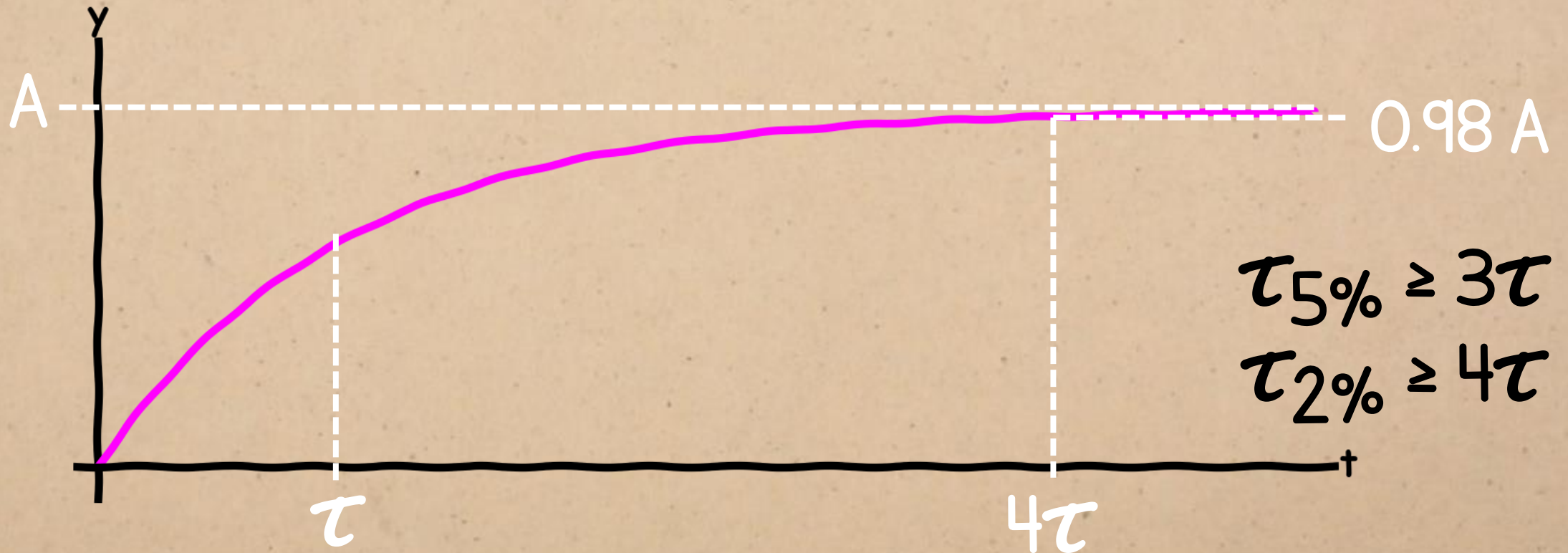
FIRST-ORDER SYSTEM

settling time = time it takes for response to get within some % of its final value.



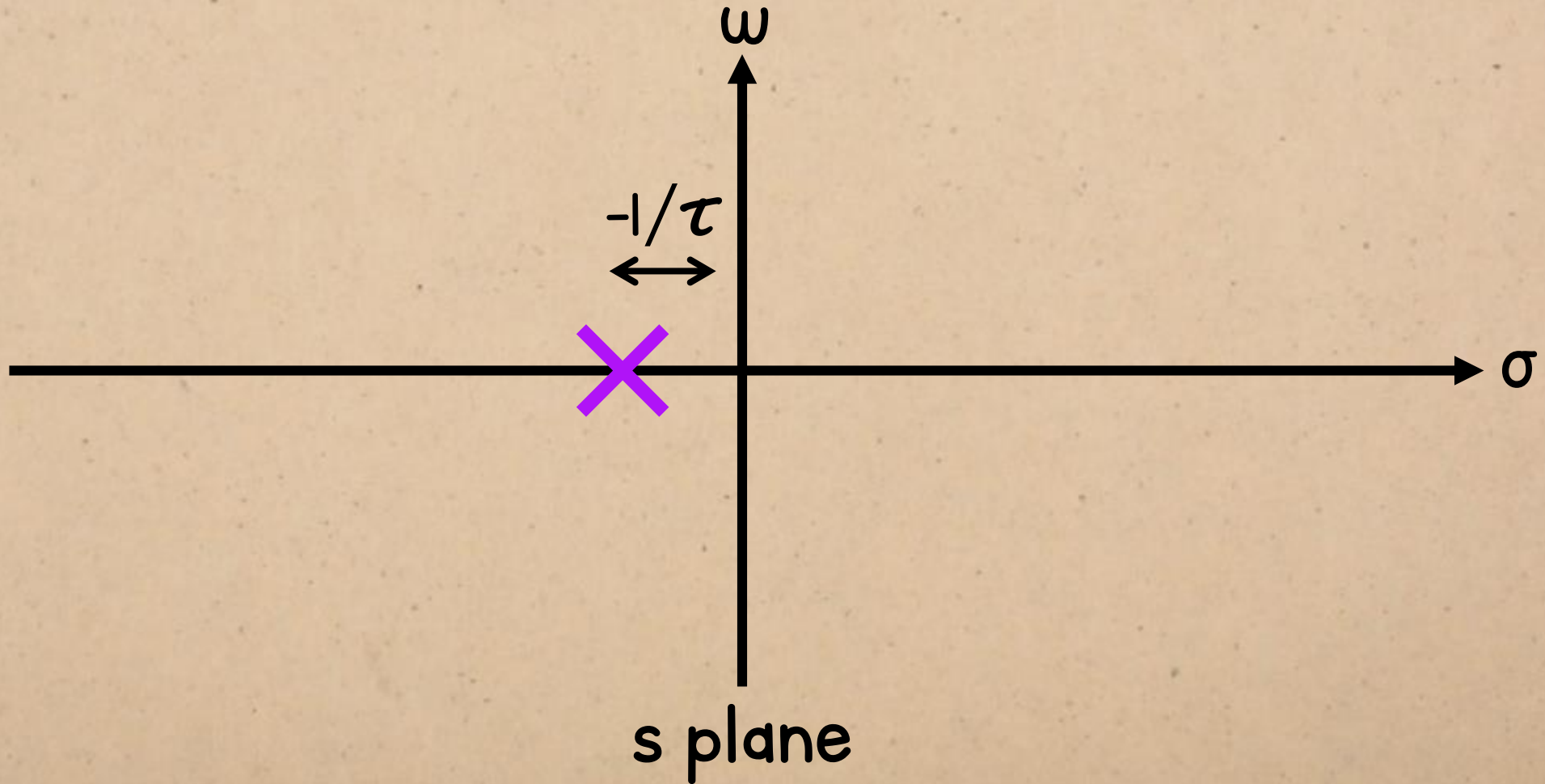
FIRST-ORDER SYSTEM

settling time = time it takes for response to get within some % of its final value.



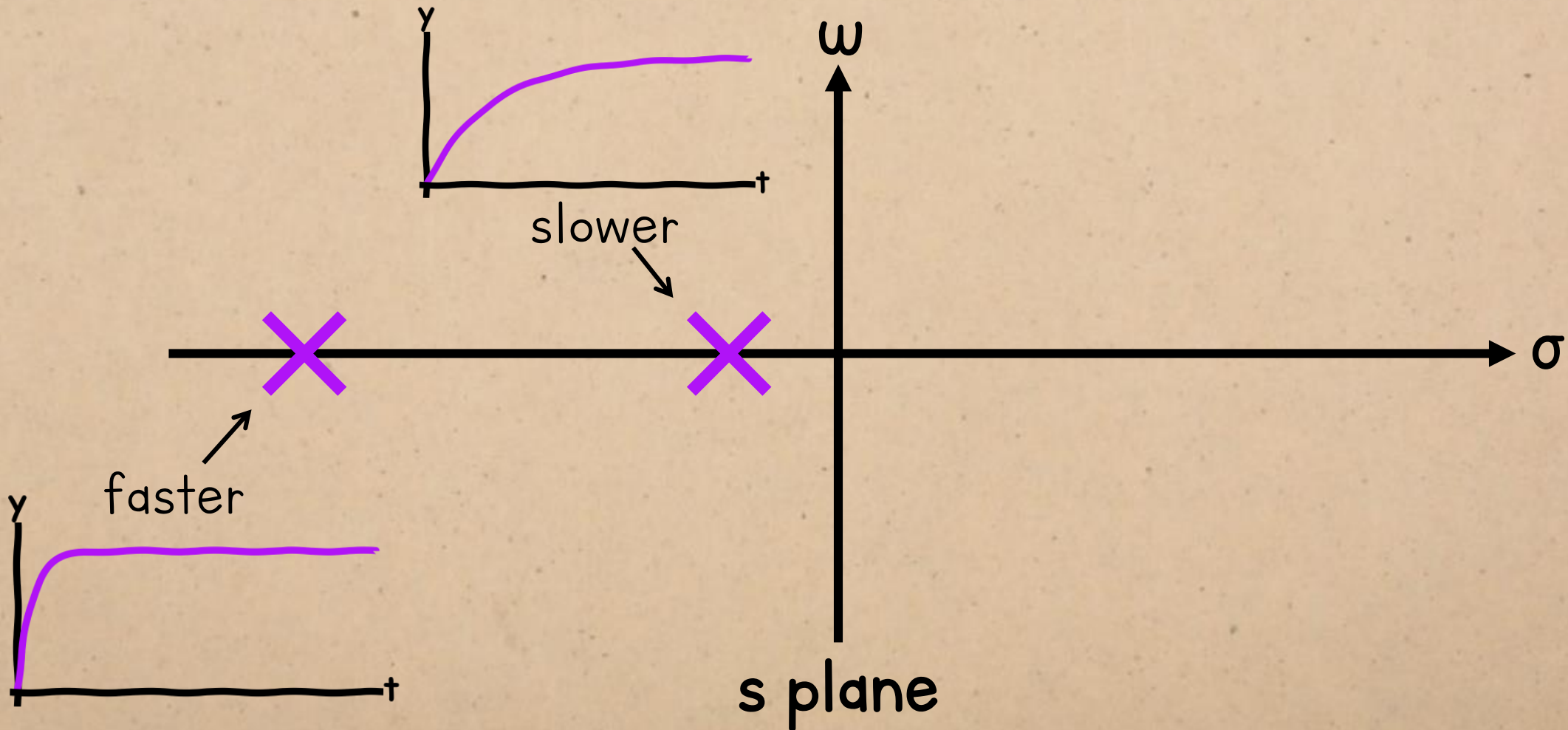
FIRST-ORDER SYSTEM

The distance from the pole to the origin determines the time constant.



FIRST-ORDER SYSTEM

The distance from the pole to the origin determines the time constant.



SECOND-ORDER SYSTEM

$$\frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

The other two types of modes we'll look at require another pole to make, so now we'll look at the general second-order model.

SECOND-ORDER SYSTEM

gain

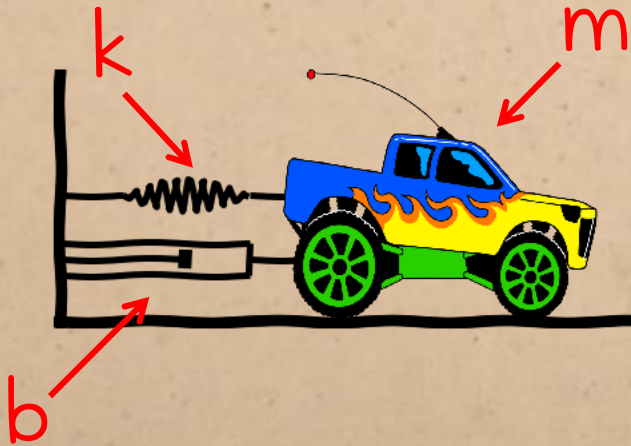
$$A\omega_n^2$$

$$s^2 + 2\zeta\omega_n s + \omega_n^2$$

SECOND-ORDER SYSTEM

The other parameters are based on the concept of a harmonic oscillator, so we'll understand what they mean better by comparing the general form to the transfer function for the spring-mass-damper system.

$$ms^2 + bs + k$$

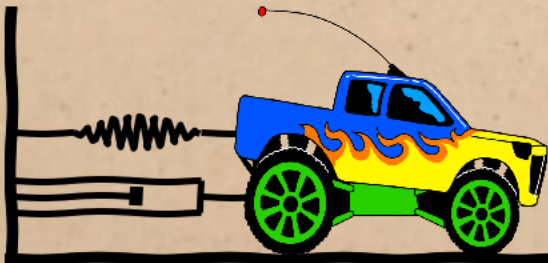


SECOND-ORDER SYSTEM

$$A\omega_n^2$$

$$s^2 + 2\zeta\omega_n s + \omega_n^2$$

natural frequency



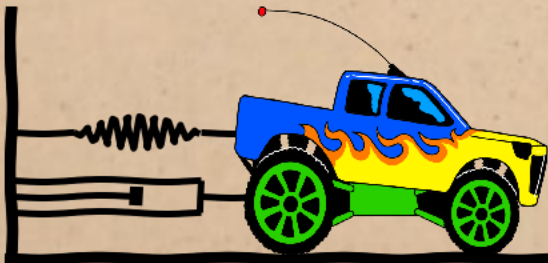
SECOND-ORDER SYSTEM

$$A\omega_n^2$$

$$\frac{1}{ms^2 + bs + k}$$

$$s^2 + 2\zeta\omega_n s + \omega_n^2$$

natural frequency



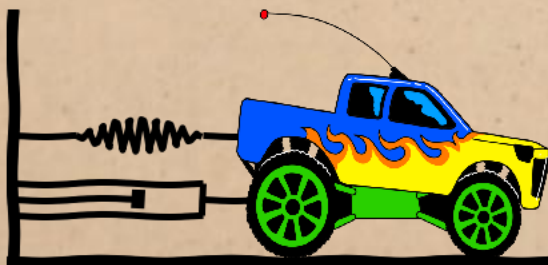
SECOND-ORDER SYSTEM

$$A\omega_n^2$$

$$\frac{1/m}{s^2 + b/m s + k/m}$$

$$s^2 + 2\zeta\omega_n s + \omega_n^2$$

natural frequency ↙



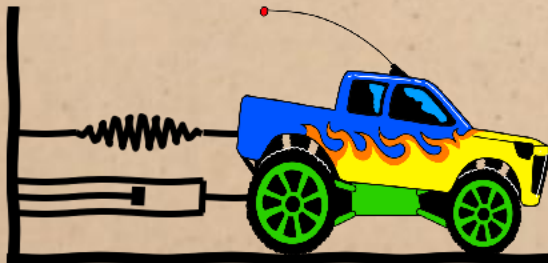
SECOND-ORDER SYSTEM

$$A\omega_n^2$$

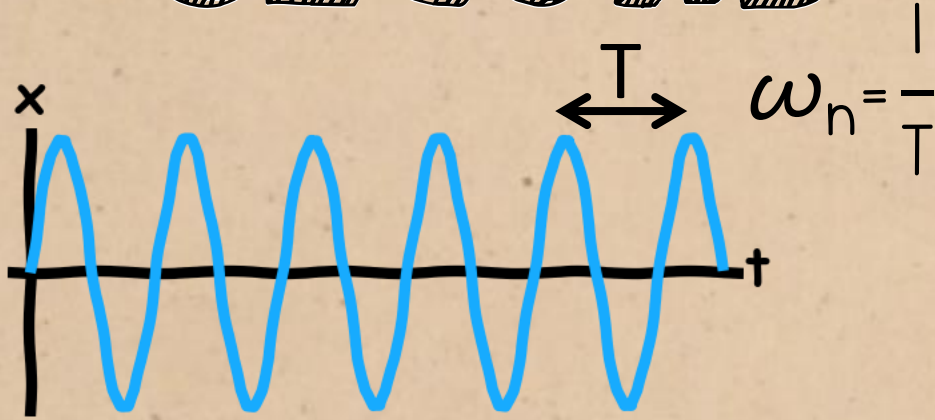
$$\frac{1/m}{s^2 + b/m s + k/m}$$

$$s^2 + 2\zeta\omega_n s + \omega_n^2$$

natural frequency
 $\omega_n = \sqrt{k/m}$



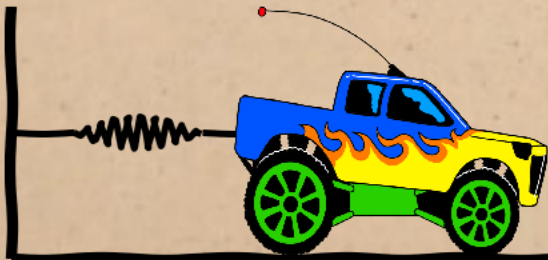
SECOND-ORDER SYSTEM



$$A\omega_n^2$$

natural frequency = frequency
the system would oscillate
without damping.

$$s^2 + 2\zeta\omega_n s + \omega_n^2$$



Common Unilateral Laplace Transform Pairs

$x(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} X(s)e^{st} ds$	$X(s) = \int_0^\infty x(t)e^{-st} dt$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
$tu(t)$	$\frac{1}{s^2}$
$t^n u(t)$	$\frac{n!}{s^{n+1}}$
$e^{\lambda t} u(t)$	$\frac{1}{s-\lambda}$
$te^{\lambda t} u(t)$	$\frac{1}{(s-\lambda)^2}$
$t^n e^{\lambda t} u(t)$	$\frac{n!}{(s-\lambda)^{n+1}}$
$\cos(bt)u(t)$	$\frac{s}{s^2+b^2}$
$\sin(bt)u(t)$	$\frac{b}{s^2+b^2}$
$e^{-at} \cos(bt)u(t)$	$\frac{s+a}{(s+a)^2+b^2}$
$e^{-at} \sin(bt)u(t)$	$\frac{b}{(s+a)^2+b^2}$
$re^{-at} \cos(bt + \theta)u(t)$	$\frac{(r \cos \theta)s + (ar \cos \theta - br \sin \theta)}{s^2 + 2as + (a^2 + b^2)}$
$re^{-at} \cos(bt + \theta)u(t)$	$\frac{0.5re^{j\theta}}{s+a-jb} + \frac{0.5re^{-j\theta}}{s+a+jb}$
$re^{-at} \cos(bt + \theta)u(t)$	$\frac{As+B}{s^2+2as+c}$
	$r = \sqrt{\frac{A^2c+B^2-2ABa}{c-a^2}}, b = \sqrt{c-a^2}, \theta = \tan^{-1} \frac{Aa-b}{A\sqrt{c-a^2}}$

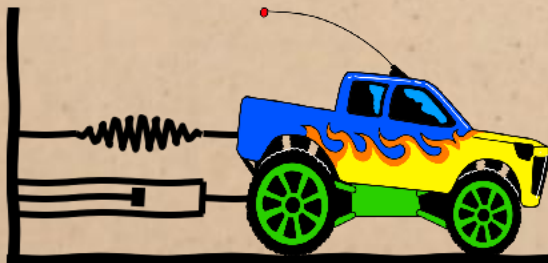
Notice that the form of the undamped response is the same as the Laplace transform for a sinusoid.

Trigonometric identities

SECOND-ORDER SYSTEM

damping ratio

$$\frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$



SECOND-ORDER SYSTEM

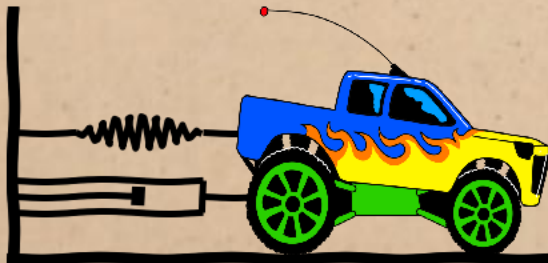
damping ratio

$$2\zeta\omega_n = \frac{b}{m}$$

$$A\omega_n^2$$

$$\frac{1/m}{s^2 + b/m s + k/m}$$

$$s^2 + 2\zeta\omega_n s + \omega_n^2$$



SECOND-ORDER SYSTEM

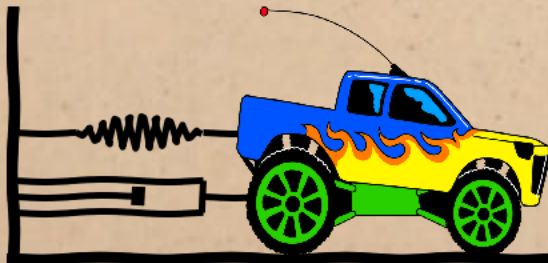
damping ratio

$$\zeta = \frac{b}{2m\omega_n}$$

$$A\omega_n^2$$

$$\frac{1/m}{s^2 + b/m s + k/m}$$

$$s^2 + 2\zeta\omega_n s + \omega_n^2$$



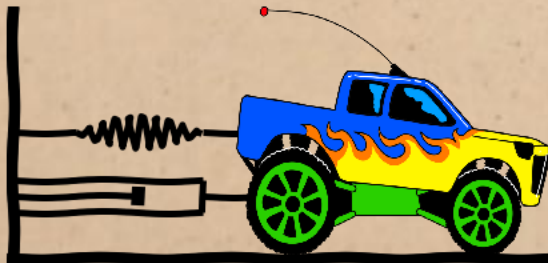
SECOND-ORDER SYSTEM

damping ratio = ratio of actual damping of the system to the critical damping value.

$$A\omega_n^2$$

This won't make a lot of sense until we understand how the damping coefficient affects the pole positions – and therefore the response characteristics.

$$s^2 + 2\zeta\omega_n s + \omega_n^2$$



...So we'll use the quadratic equation to find the poles of the second-order system in terms of these parameters.

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

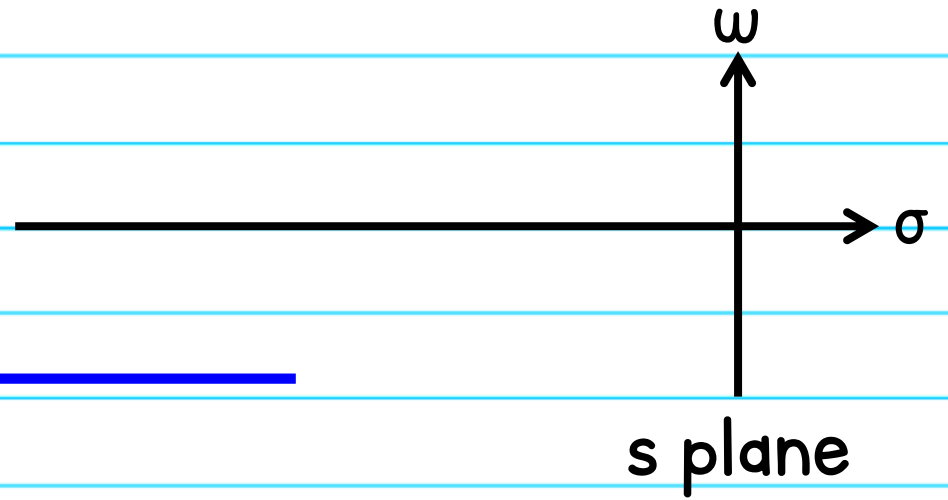
$$ms^2 + bs + k$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$ms^2 + bs + k$$

$$\frac{-b \pm \sqrt{b^2 - 4mk}}{2m}$$


$$\frac{-b \pm \sqrt{b^2 - 4mk}}{2m}$$



STABILITY

Positive pole if:

★ m or k is negative



$$\frac{-b \pm \sqrt{b^2 - 4mk}}{2m}$$

$$\sqrt{b^2 - 4mk} > b$$
$$\therefore b^2 - 4mk > b^2$$

$$\therefore 4mk < 0$$

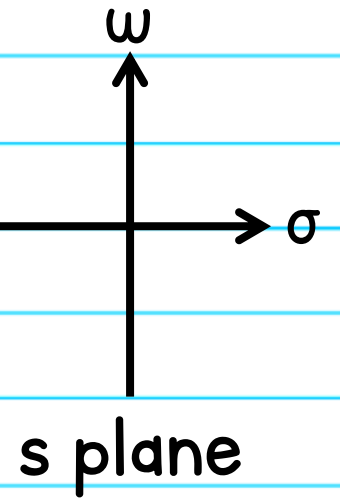
STABILITY

Positive pole if:

~~★ m or k is negative~~

But that doesn't make
physical sense

$$\frac{-b \pm \sqrt{b^2 - 4mk}}{2m}$$



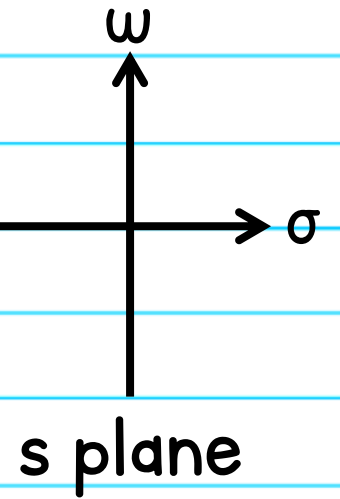
STABILITY

Positive pole if:

~~★ m or k is negative~~

★ b is negative

$$\frac{-(-b) \pm \sqrt{b^2 - 4mk}}{2m}$$



We usually think of the damping coefficient as being positive, but the concept of a negative damper gives us insight about what it means for a system to be stable or unstable, from the perspective of energy.

STABILITY

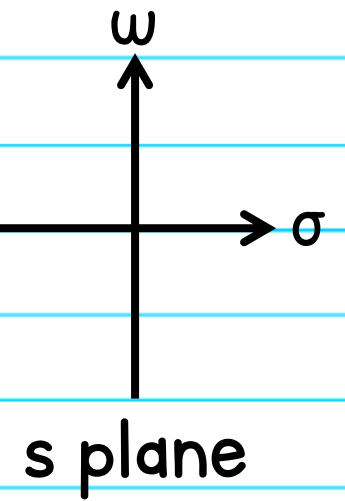
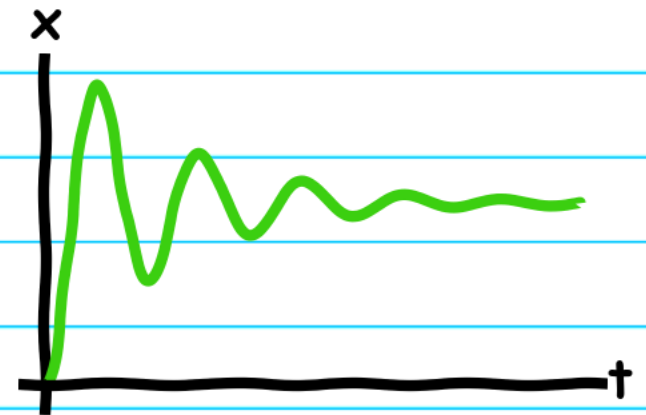
Positive pole if:

~~★ m or k is negative~~

★ b is negative

$$\frac{-b \pm \sqrt{b^2 - 4mk}}{2m}$$

dampers dissipate energy



STABILITY

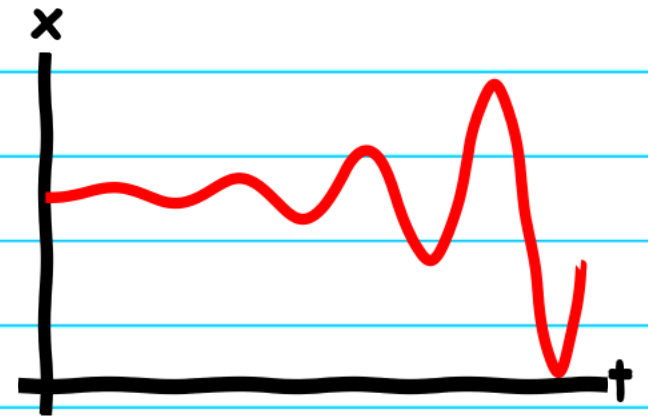
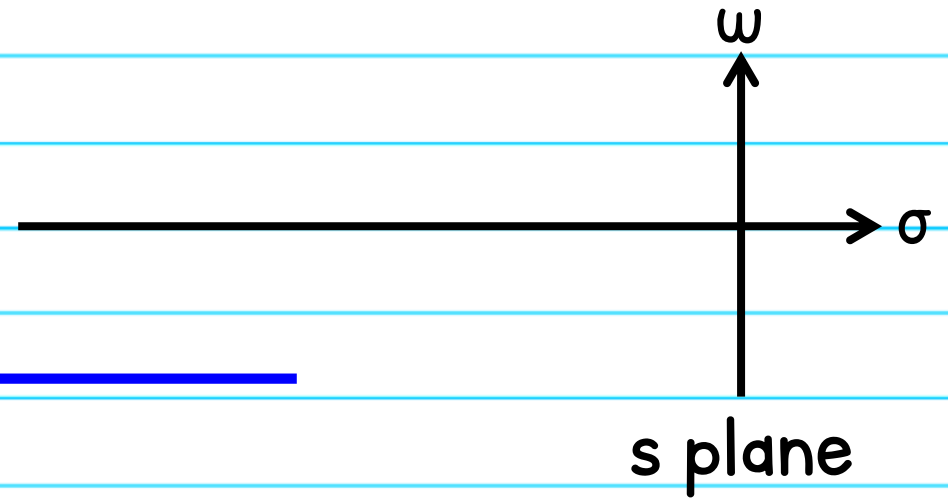
Positive pole if:

~~★ m or k is negative~~

★ b is negative

$$\frac{b \pm \sqrt{b^2 - 4mk}}{2m}$$

dampers dissipate energy
 $\therefore$ damper with $-b$ adds energy



STABILITY

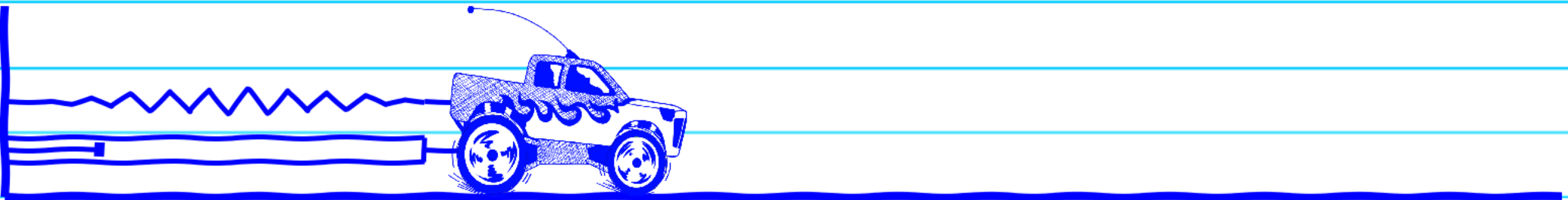
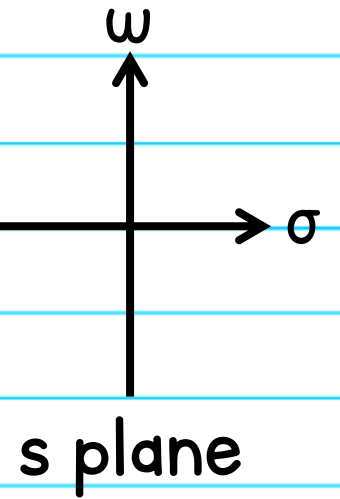
Positive pole if:

~~★ m or k is negative~~

★ b is negative

$$\frac{-b \pm \sqrt{b^2 - 4mk}}{2m}$$

stable systems lose energy



STABILITY

Positive pole if:

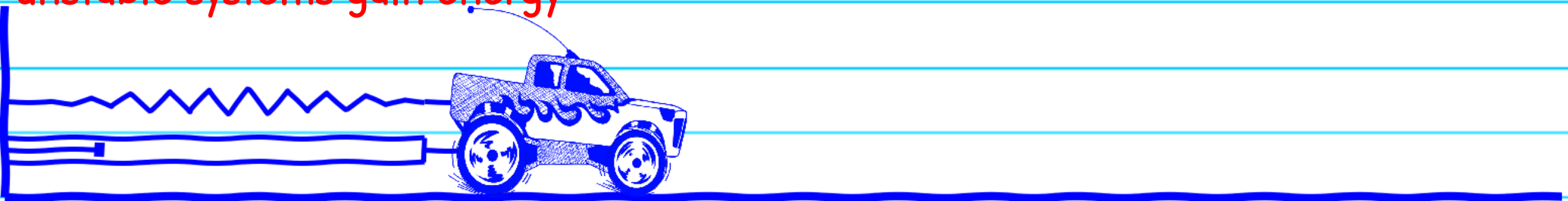
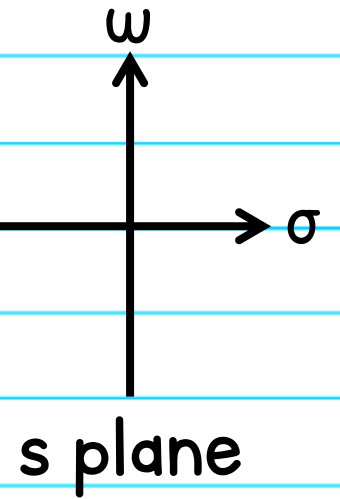
~~★ m or k is negative~~

★ b is negative

$$\frac{-b \pm \sqrt{b^2 - 4mk}}{2m}$$

stable systems lose energy

unstable systems gain energy



STABILITY

Positive pole if:

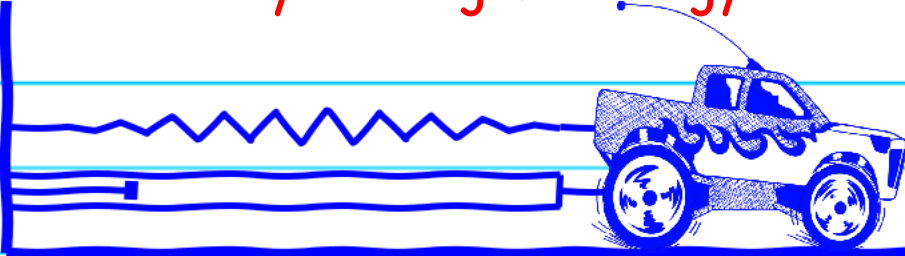
~~★ m or k is negative~~

~~★ b is negative~~

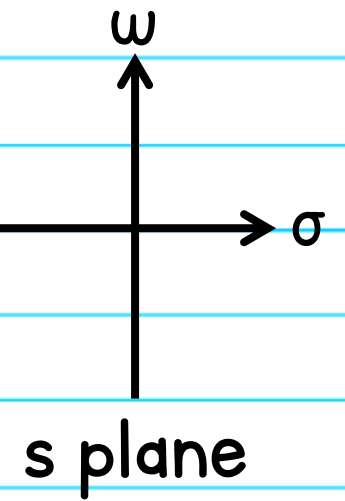
$$\frac{-b \pm \sqrt{b^2 - 4mk}}{2m}$$

stable systems lose energy

unstable systems gain energy



So as long as the system is damped, it should be stable.



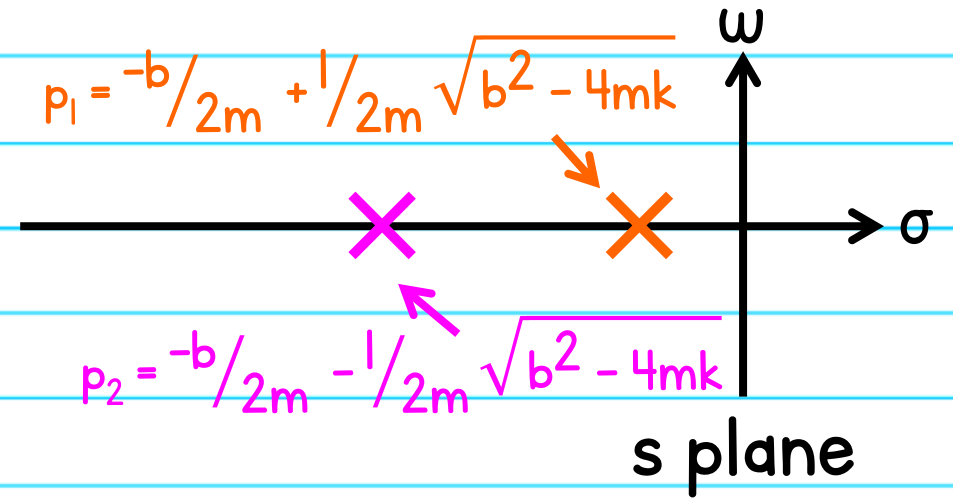
FORM OF RESPONSE

Depending on the value under the square root, there are three possible forms for the second-order response.

$$\frac{-b \pm \sqrt{b^2 - 4mk}}{2m}$$

FORM OF RESPONSE

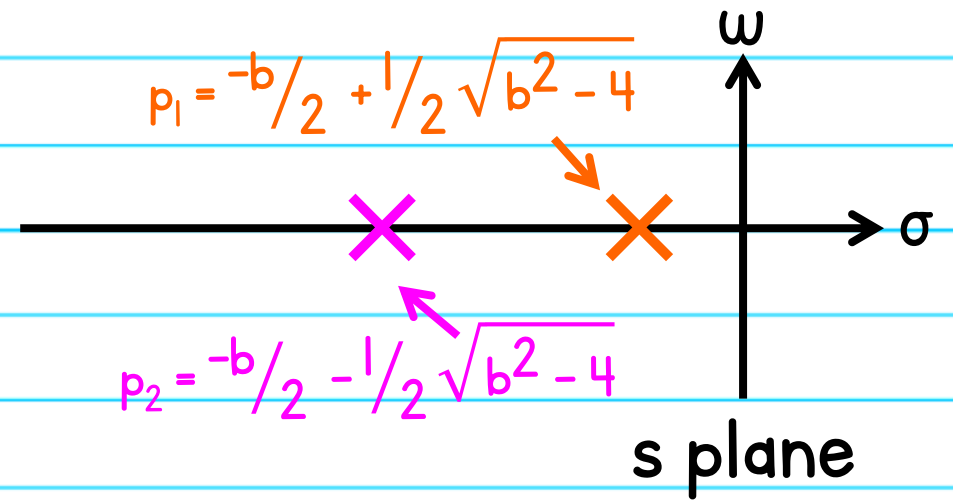
☆ $b^2 - 4mk > 0 \rightarrow$ two real poles



$$b^2 - 4mk > 0$$

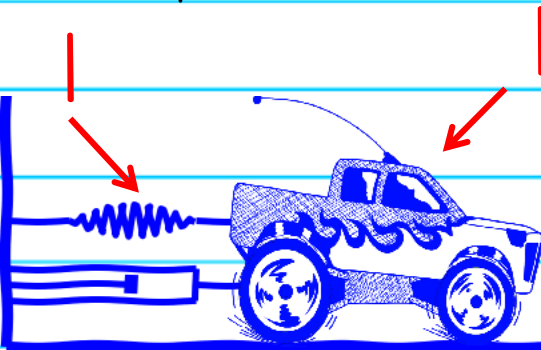
FORM OF RESPONSE

☆ $b^2 - 4mk > 0 \rightarrow$ two real poles



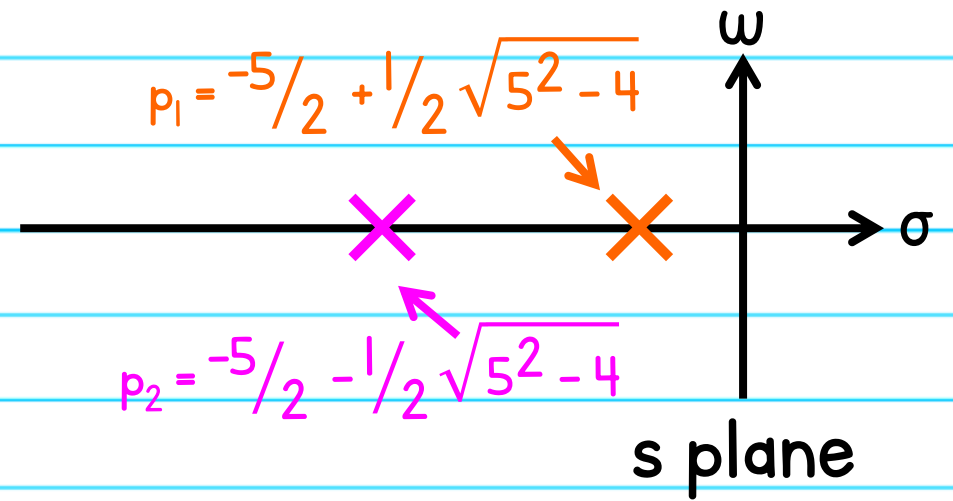
$$b^2 - 4 > 0$$

Example:



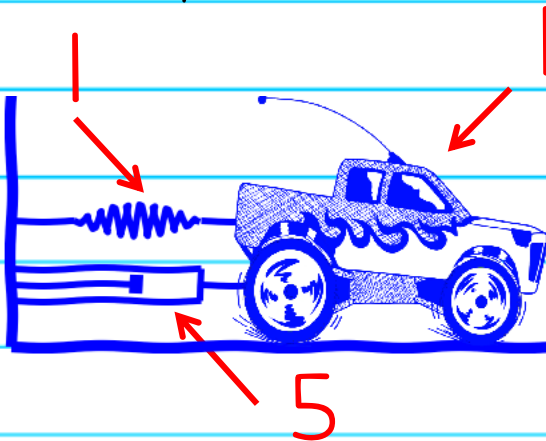
FORM OF RESPONSE

☆ $b^2 - 4mk > 0 \rightarrow$ two real poles



$$b^2 - 4 > 0$$

The system will have real poles if $b > 2$, so we let $b = 5$

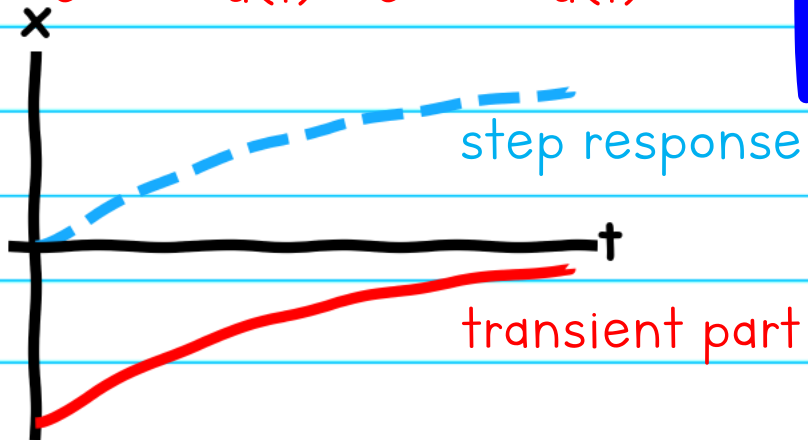


FORM OF RESPONSE

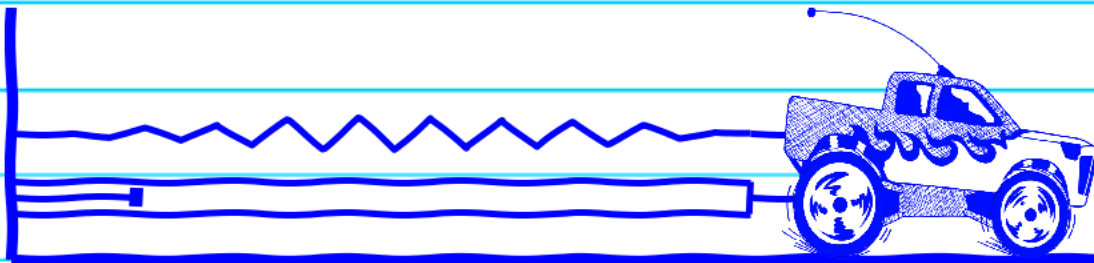
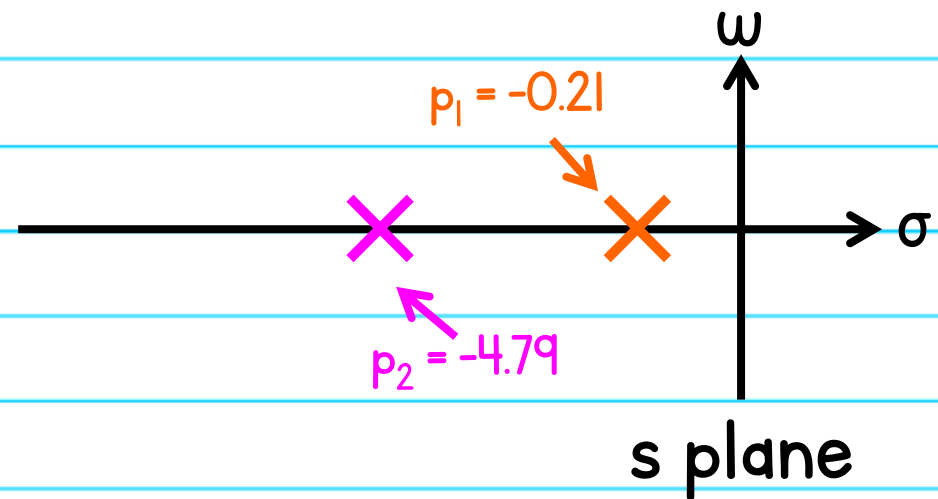
☆ $b^2 - 4mk > 0 \rightarrow$ two real poles

The response is the sum of two first-order exponential modes.

$$-e^{-0.21t}u(t) + e^{-4.79t}u(t)$$



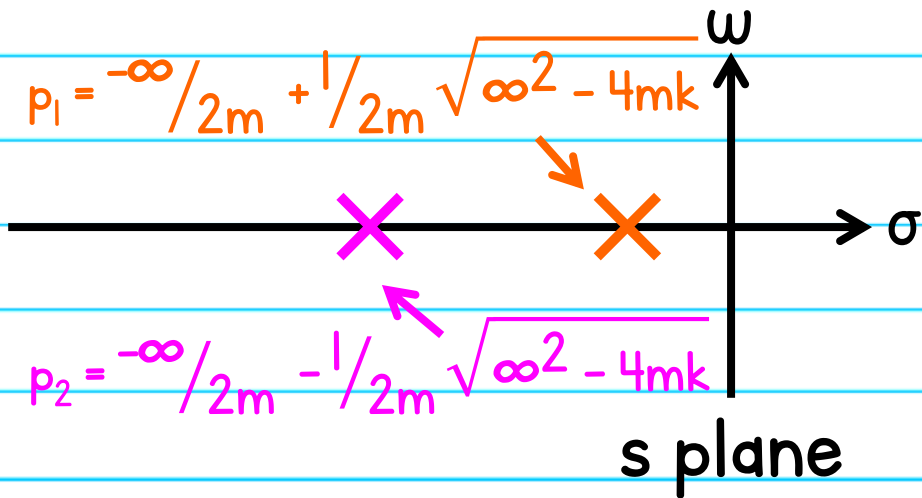
$$b^2 - 4 > 0$$



FORM OF RESPONSE

☆ $b^2 - 4mk > 0 \rightarrow$ two real poles

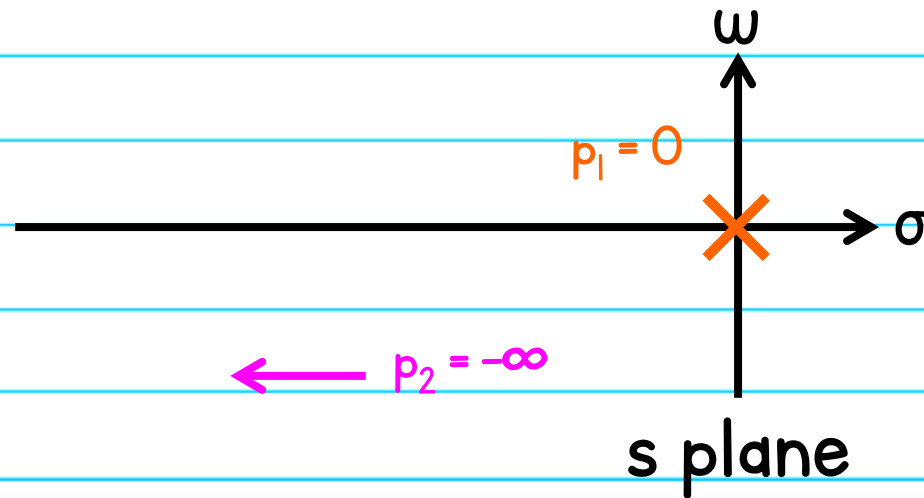
increasing $b \rightarrow \infty$



FORM OF RESPONSE

☆ $b^2 - 4mk > 0 \rightarrow$ two real poles

increasing $b \rightarrow \infty$

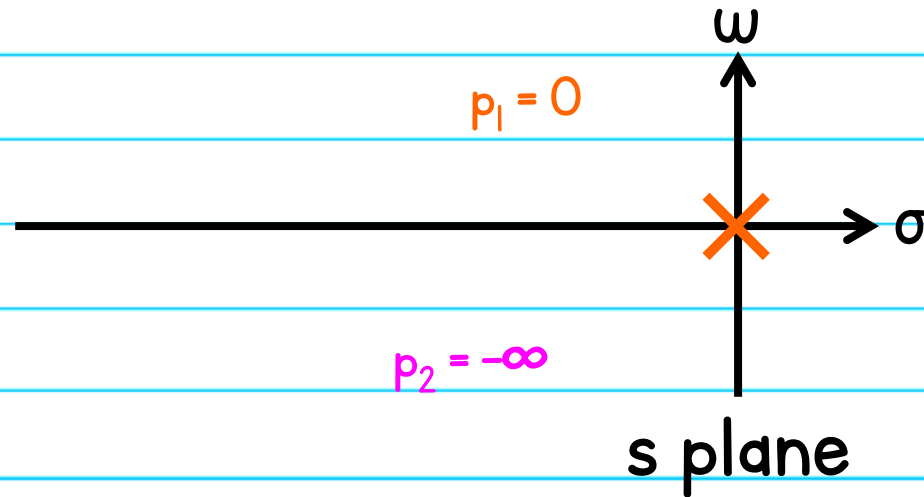


As the damping coefficient increases, one of the poles approaches zero, while the other approaches negative infinity.

FORM OF RESPONSE

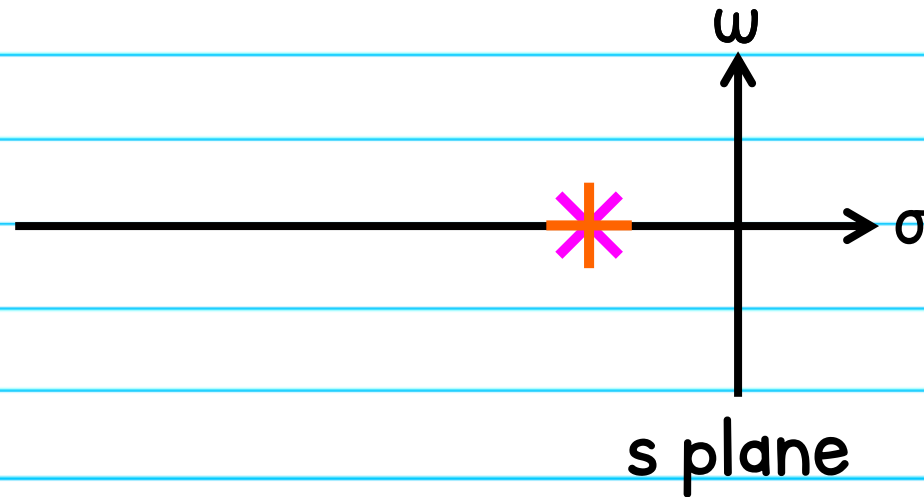
☆ $b^2 - 4mk > 0 \rightarrow$ two real poles

The settling time is only as fast as the slowest pole, so
increasing b = increasing settling time



FORM OF RESPONSE

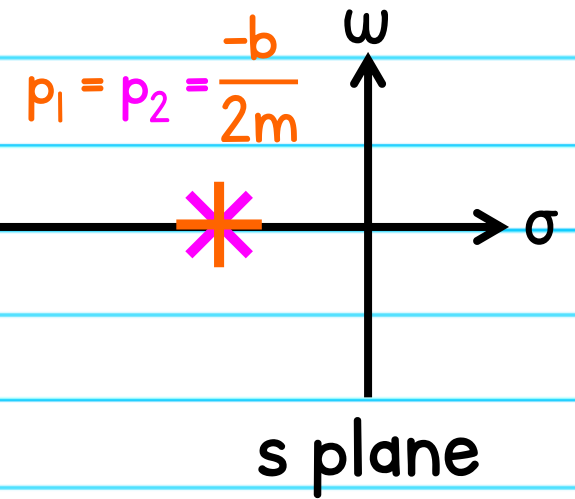
☆ $b^2 - 4mk > 0 \rightarrow$ two real poles



The fastest possible response happens
when the two poles coincide.

FORM OF RESPONSE

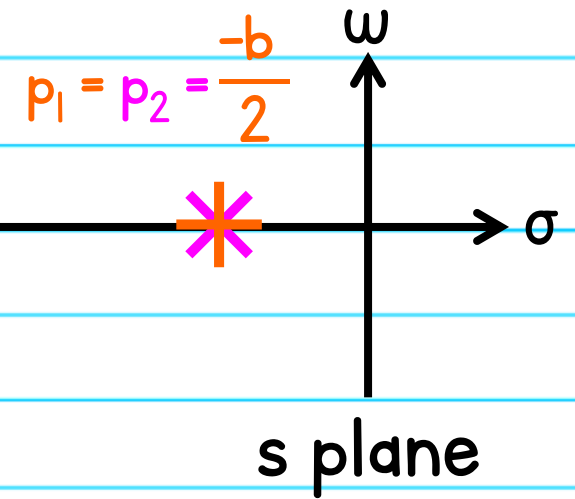
- ☆ $b^2 - 4mk > 0 \rightarrow$ two real poles
- ☆ $b^2 - 4mk = 0 \rightarrow$ repeated real pole



$$b^2 - 4mk = 0$$

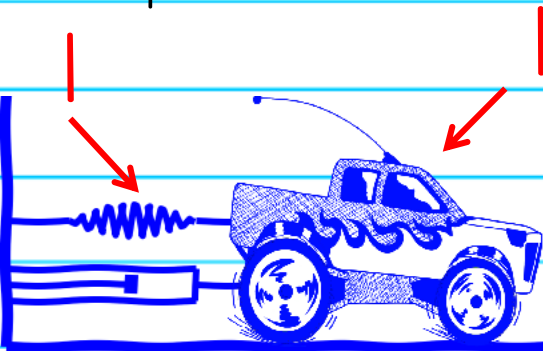
FORM OF RESPONSE

- ☆ $b^2 - 4mk > 0 \rightarrow$ two real poles
- ☆ $b^2 - 4mk = 0 \rightarrow$ repeated real pole



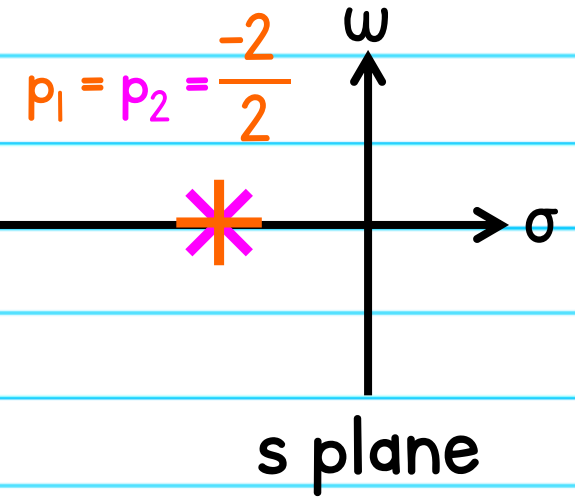
$$b^2 - 4mk = 0$$

Example:



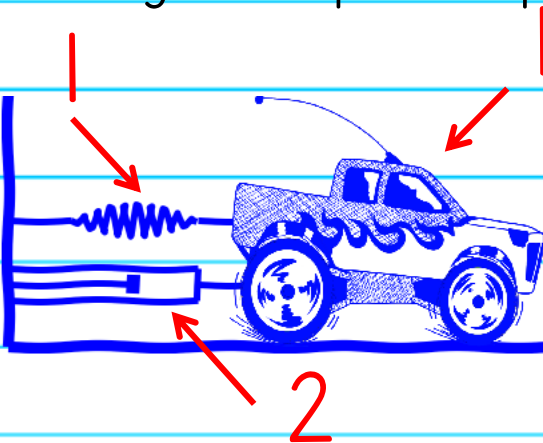
FORM OF RESPONSE

- ☆ $b^2 - 4mk > 0 \rightarrow$ two real poles
- ☆ $b^2 - 4mk = 0 \rightarrow$ repeated real pole



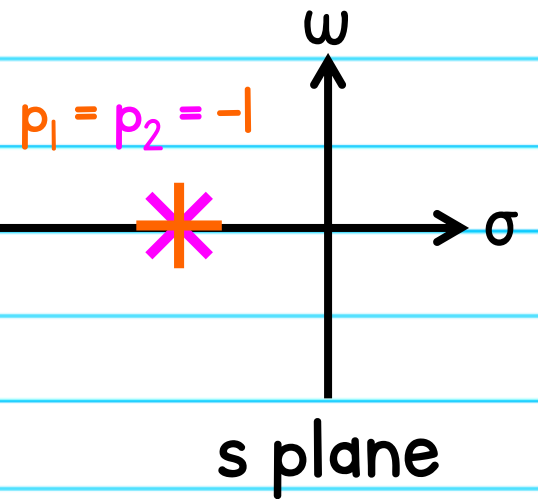
$$b^2 - 4 = 0$$

We get a repeated pole when $b = 2$



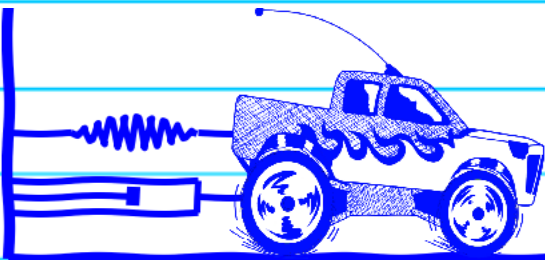
FORM OF RESPONSE

- ☆ $b^2 - 4mk > 0 \rightarrow$ two real poles
- ☆ $b^2 - 4mk = 0 \rightarrow$ repeated real pole



$$b^2 - 4mk = 0$$

$$x(s) = \frac{1}{(s+1)^2}$$



Time convolution

$$\int_{-\infty}^c x(\lambda) d\lambda \iff \frac{1}{s} X(s) + \frac{1}{s} \int_{-\infty}^c x(\lambda) d\lambda$$

Frequency convolution

$$x(t) * v(t) \iff X(s)V(s)$$

$$x(t)v(t) \iff \frac{1}{2\pi j} X(\omega) * V(\omega)$$

Initial value: $f(0^+) = \lim_{s \rightarrow \infty} sF(s)$

Final value: $f(\infty) = \lim_{s \rightarrow 0} sF(s)$ with all poles in left-hand plane

Common Unilateral Laplace Transform Pairs

$x(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} X(s)e^{st} ds$	$X(s) = \int_0^\infty x(t)e^{-st} dt$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
$tu(t)$	$\frac{1}{s^2}$
$t^n u(t)$	$\frac{n!}{s^{n+1}}$
$e^{\lambda t} u(t)$	$\frac{1}{s-\lambda}$
$te^{\lambda t} u(t)$	$\frac{1}{(s-\lambda)^2}$
$t^n e^{\lambda t} u(t)$	$\frac{n!}{(s-\lambda)^{n+1}}$
$\cos(bt)u(t)$	$\frac{s}{s^2+b^2}$
$\sin(bt)u(t)$	$\frac{b}{s^2+b^2}$
$e^{-at} \cos(bt)u(t)$	$\frac{s+a}{(s+a)^2+b^2}$
$e^{-at} \sin(bt)u(t)$	$\frac{b}{(s+a)^2+b^2}$
$re^{-at} \cos(bt + \theta)u(t)$	$\frac{(r \cos \theta)s + (ar \cos \theta - br \sin \theta)}{s^2 + 2as + (a^2 + b^2)}$
$re^{-at} \cos(bt + \theta)u(t)$	$\frac{0.5re^{j\theta}}{s+a-jb} + \frac{0.5re^{-j\theta}}{s+a+jb}$
$re^{-at} \cos(bt + \theta)u(t)$	$\frac{As+B}{s^2+2as+c}$
	$r = \sqrt{\frac{A^2c+B^2-2ABa}{c-a^2}}, b = \sqrt{c-a^2}, \theta = \tan^{-1} \frac{Aa-b}{A\sqrt{c-a^2}}$

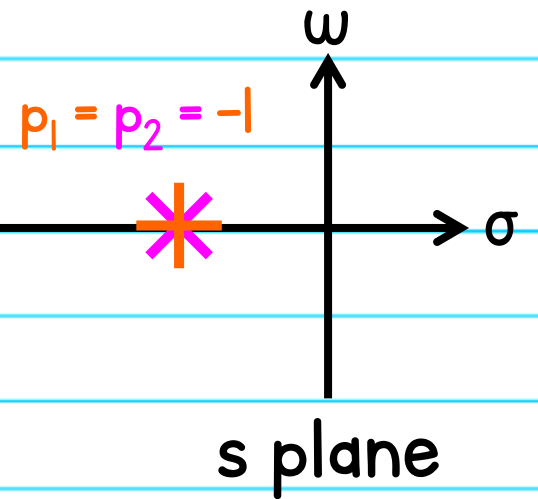
If a pole is repeated, its exponential response in the time domain gets multiplied by t.

(or some power of t, if there's more than two poles in the same spot)

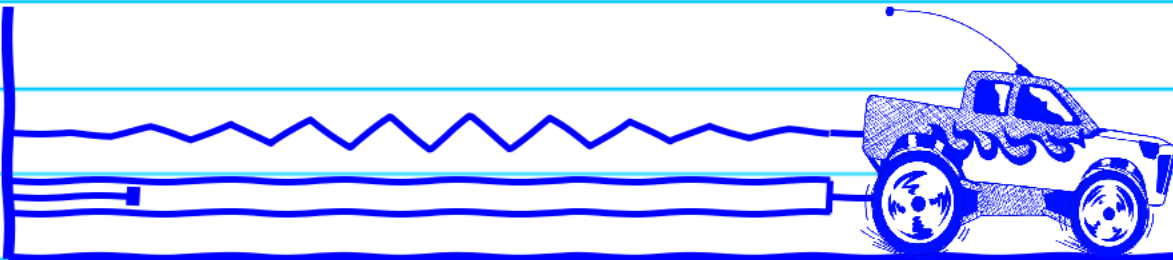
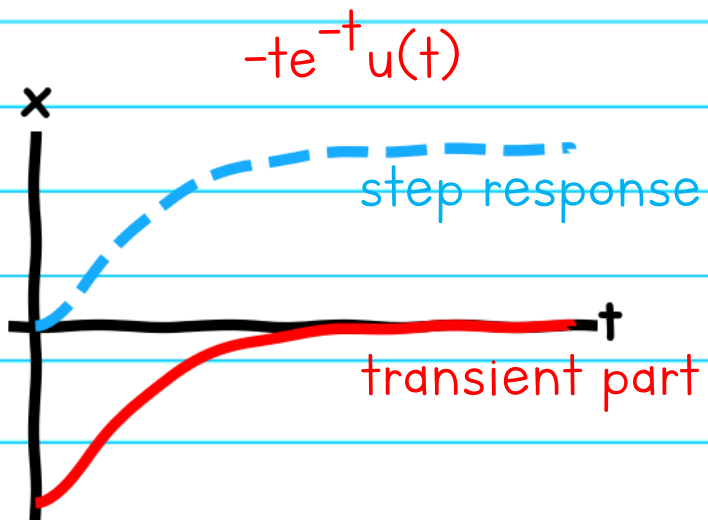
Trigonometric identities

FORM OF RESPONSE

- ☆ $b^2 - 4mk > 0 \rightarrow$ two real poles
- ☆ $b^2 - 4mk = 0 \rightarrow$ repeated real pole

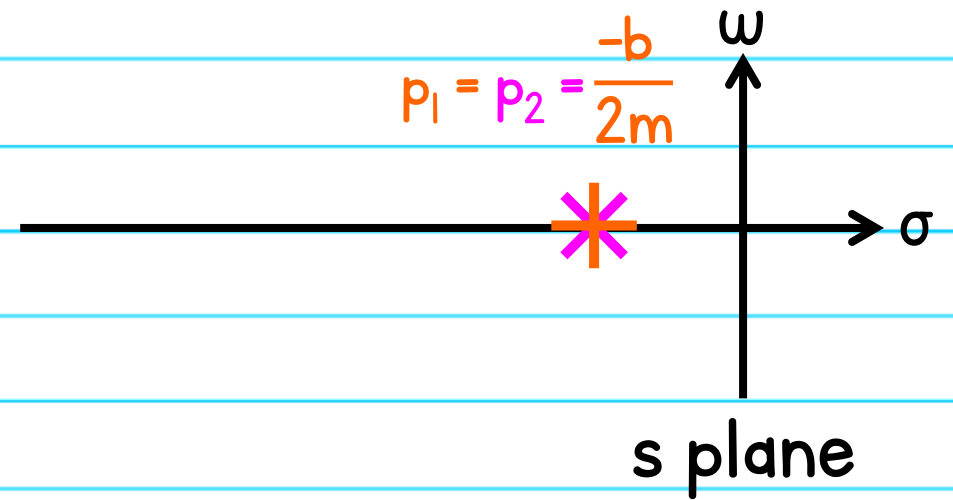


$$b^2 - 4 = 0$$



FORM OF RESPONSE

- ☆ $b^2 - 4mk > 0 \rightarrow$ two real poles
- ☆ $b^2 - 4mk = 0 \rightarrow$ repeated real pole

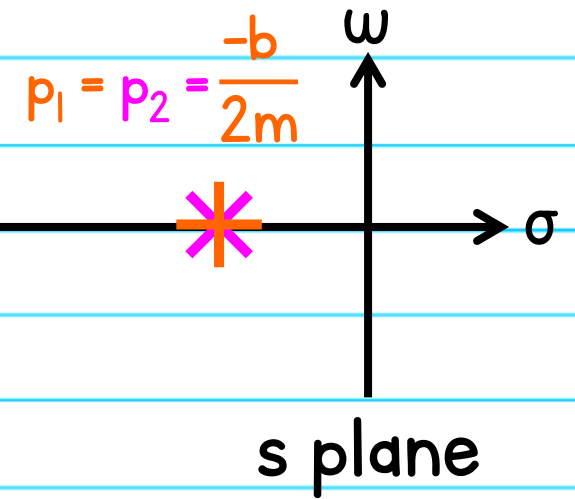


$$b^2 - 4mk = 0$$
$$\therefore b^2 = 4mk$$

$$\omega_n = \sqrt{k/m}$$

FORM OF RESPONSE

- ☆ $b^2 - 4mk > 0 \rightarrow$ two real poles
- ☆ $b^2 - 4mk = 0 \rightarrow$ repeated real pole



$$b^2 - 4mk = 0$$

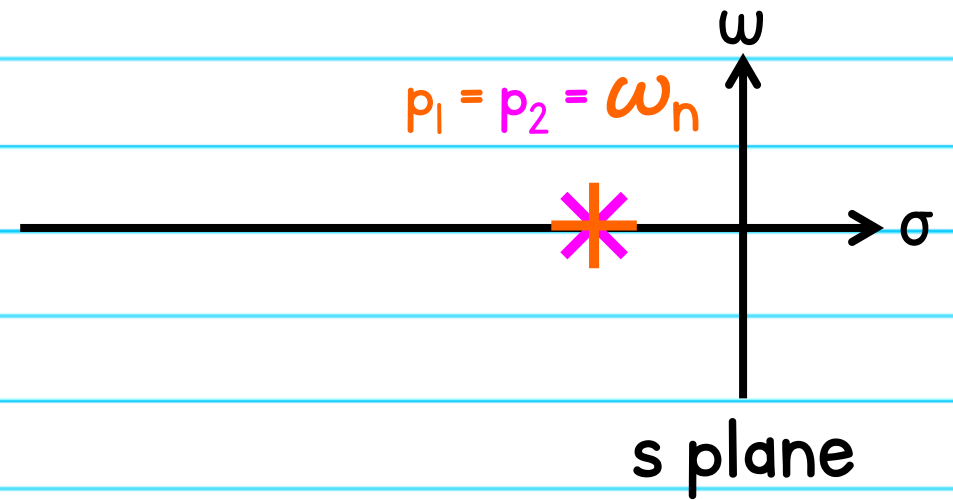
$$\therefore b^2 = 4mk$$

$$\therefore b = 2\sqrt{mk}$$

$$\omega_n = \sqrt{k/m}$$

FORM OF RESPONSE

- ☆ $b^2 - 4mk > 0 \rightarrow$ two real poles
- ☆ $b^2 - 4mk = 0 \rightarrow$ repeated real pole



$$b^2 - 4mk = 0$$

$$\therefore b^2 = 4mk$$

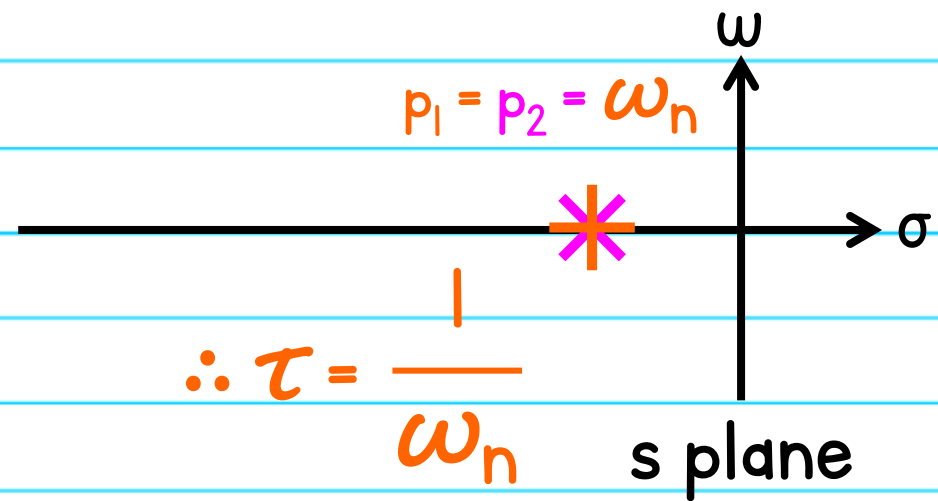
$$\therefore b = 2\sqrt{mk}$$

$$\therefore b = 2m\omega_n$$

$$\omega_n = \sqrt{k/m}$$

FORM OF RESPONSE

- ☆ $b^2 - 4mk > 0 \rightarrow$ two real poles
- ☆ $b^2 - 4mk = 0 \rightarrow$ repeated real pole



$$b^2 - 4mk = 0$$

$$\therefore b^2 = 4mk$$

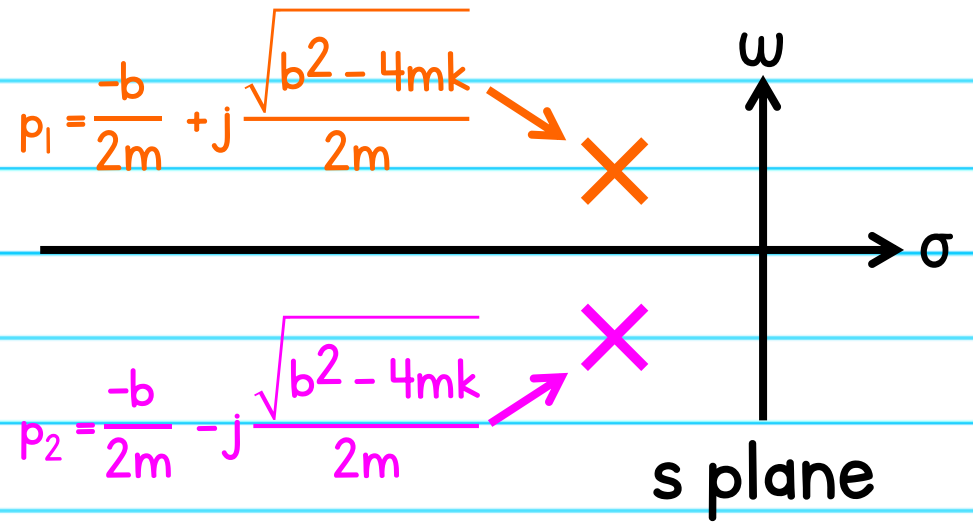
$$\therefore b = 2\sqrt{mk}$$

$$\therefore b = 2m\omega_n$$

$$\omega_n = \sqrt{k/m}$$

FORM OF RESPONSE

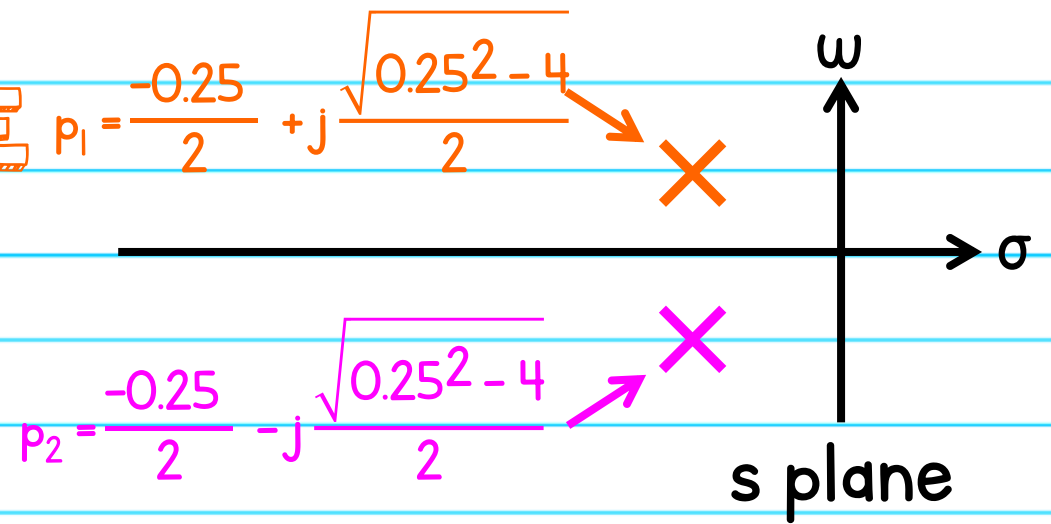
- ★ $b^2 - 4mk > 0 \rightarrow$ two real poles
- ★ $b = 2m\omega_n \rightarrow$ repeated real pole
- ★ $b^2 - 4mk < 0 \rightarrow$ complex pole pair



$$b^2 - 4mk < 0$$

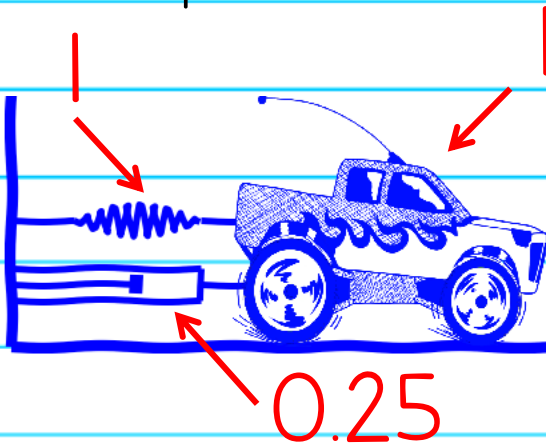
FORM OF RESPONSE

- ☆ $b^2 - 4mk > 0 \rightarrow$ two real poles
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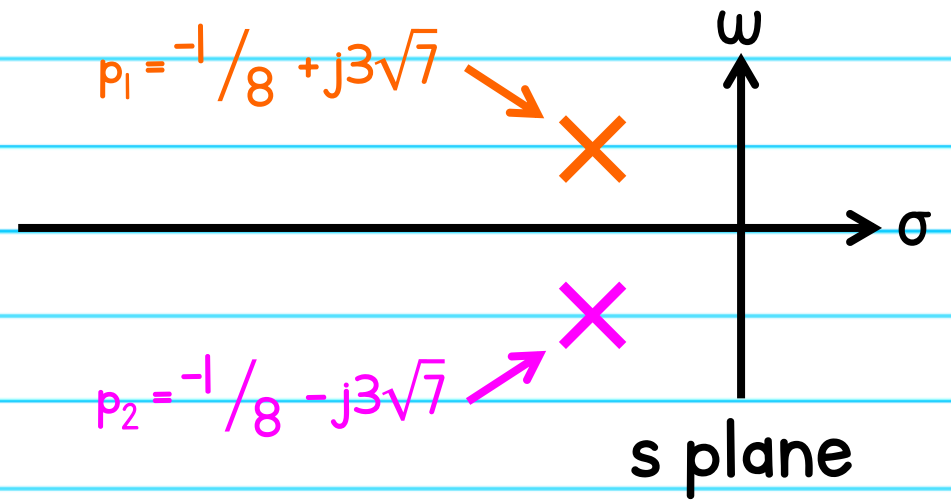
$$b^2 - 4 < 0$$

Example:

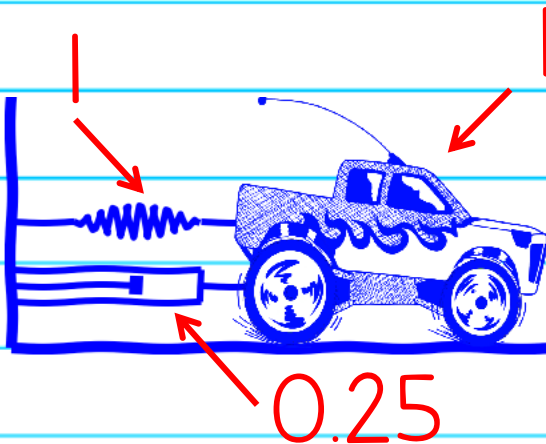


FORM OF RESPONSE

- ☆ $b^2 - 4mk > 0 \rightarrow$ two real poles
- ☆ $b = 2m\omega_n \rightarrow$ repeated real pole
- ☆ $b^2 - 4mk < 0 \rightarrow$ complex pole pair

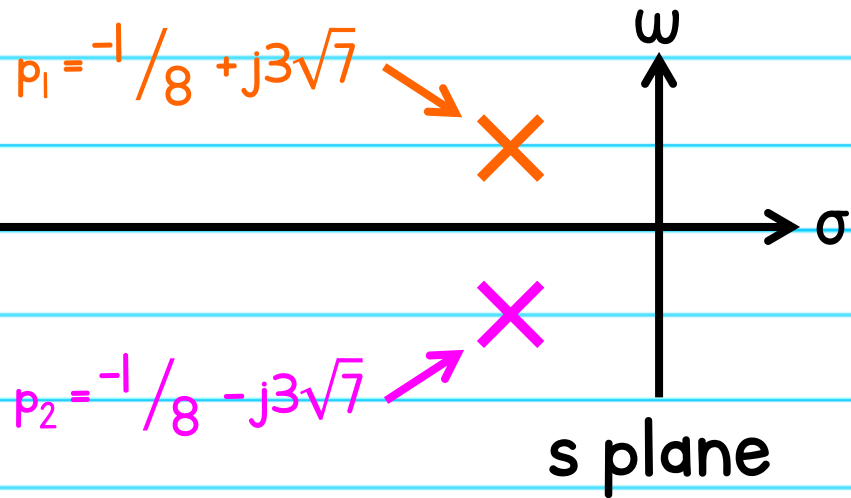


$$b^2 - 4 < 0$$



FORM OF RESPONSE

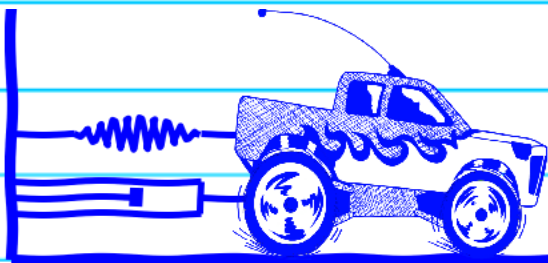
- ★ $b^2 - 4mk > 0 \rightarrow$ two real poles
- ★ $b = 2m\omega_n \rightarrow$ repeated real pole
- ★ $b^2 - 4mk < 0 \rightarrow$ complex pole pair



$$b^2 - 4 < 0$$

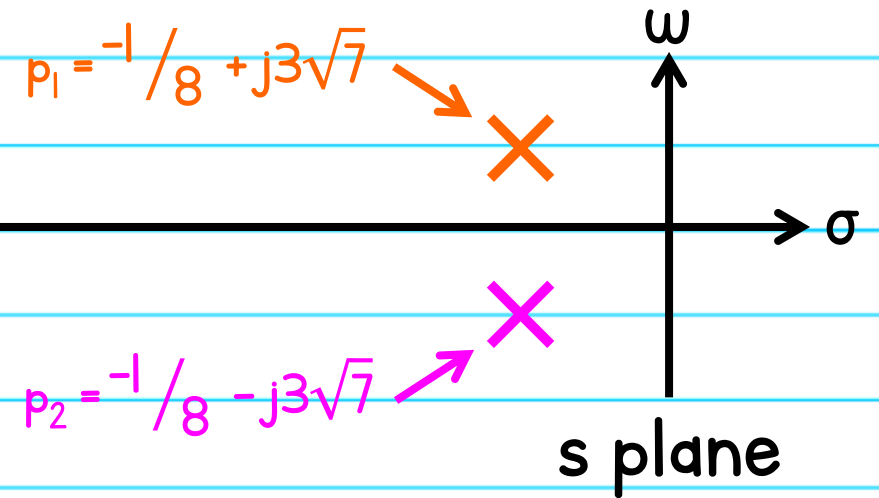
The complex poles will
always be a conjugate pair...

$$x(t) = e^{(-1/8 + j3\sqrt{7})t} + e^{(-1/8 - j3\sqrt{7})t} u(t)$$



FORM OF RESPONSE

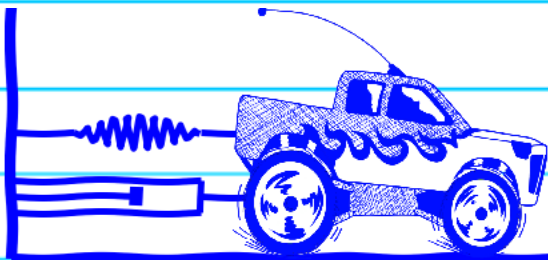
- ★ $b^2 - 4mk > 0 \rightarrow$ two real poles
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- ★ $b^2 - 4mk < 0 \rightarrow$ complex pole pair



$$b^2 - 4 < 0$$

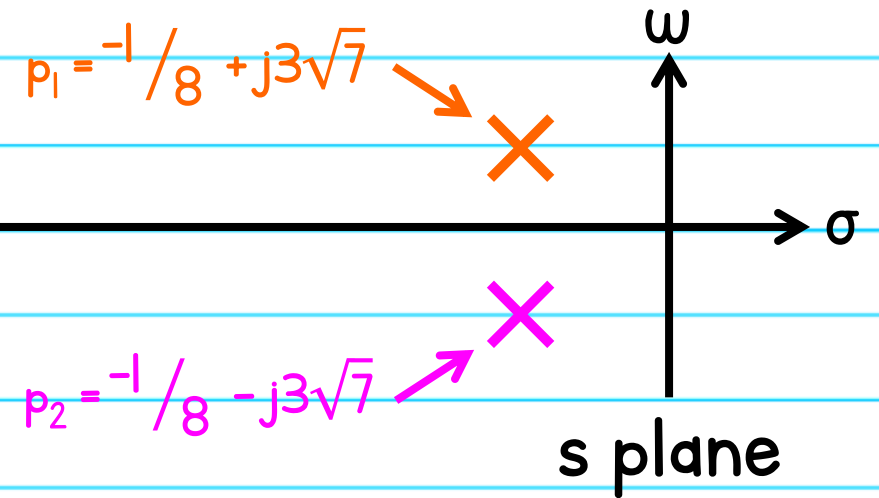
The complex poles will
always be a conjugate pair...

$$\begin{aligned} x(t) &= e^{(-1/8 + j3\sqrt{7})t} + e^{(-1/8 - j3\sqrt{7})t} u(t) \\ &= e^{-1/8t} (e^{j3\sqrt{7}t} + e^{-j3\sqrt{7}t}) u(t) \end{aligned}$$



FORM OF RESPONSE

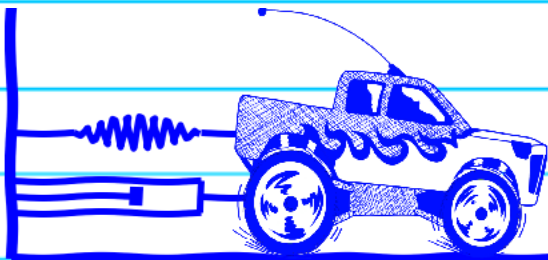
- ★ $b^2 - 4mk > 0 \rightarrow$ two real poles
- ★ $b = 2m\omega_n \rightarrow$ repeated real pole
- ★ $b^2 - 4mk < 0 \rightarrow$ complex pole pair



$$b^2 - 4mk < 0$$

So you still get a real-valued sinusoid when their responses are combined.

$$\begin{aligned} x(t) &= e^{(-1/8 + j3\sqrt{7})t} + e^{(-1/8 - j3\sqrt{7})t} u(t) \\ &= e^{-1/8t} (e^{j3\sqrt{7}t} + e^{-j3\sqrt{7}t}) u(t) \\ &= e^{-1/8t} \cos(3\sqrt{7}t) u(t) \end{aligned}$$



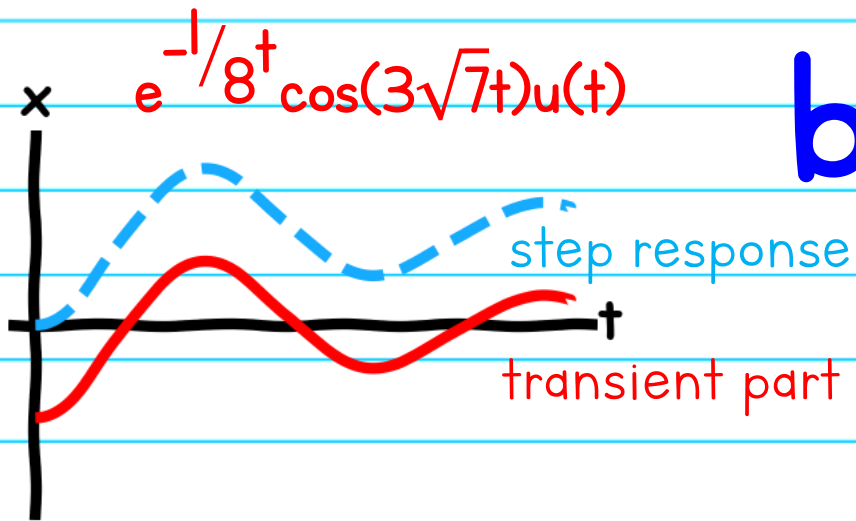
FORM OF RESPONSE

- ★ $b^2 - 4mk > 0 \rightarrow$ two real poles
- ★ $b = 2m\omega_n \rightarrow$ repeated real pole
- ★ $b^2 - 4mk < 0 \rightarrow$ complex pole pair

$$p_1 = -1/8 + j3\sqrt{7}$$

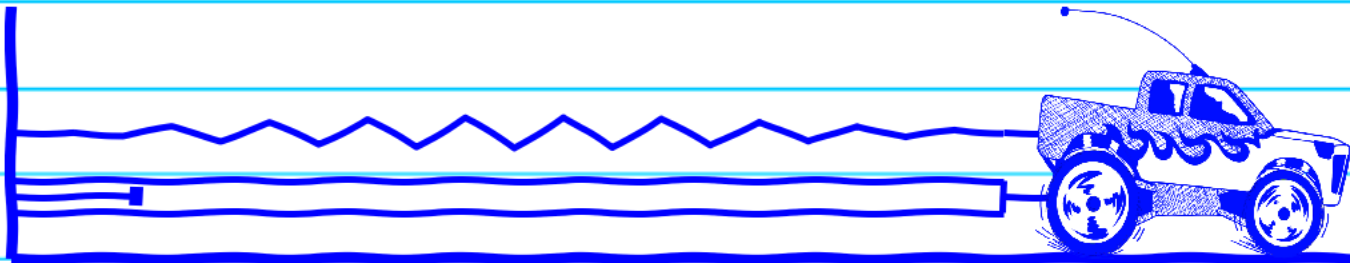
$$p_2 = -1/8 - j3\sqrt{7}$$

s plane



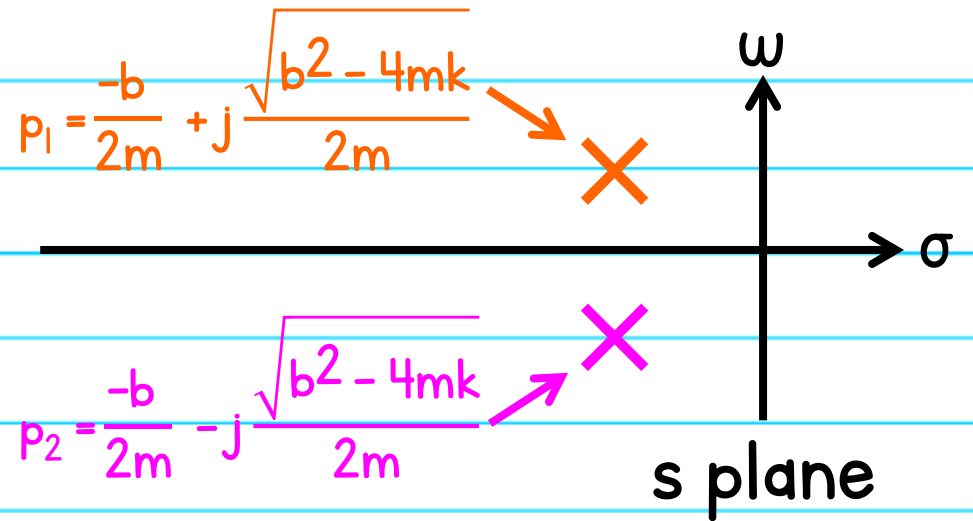
$$b^2 - 4 < 0$$

Complex poles produce oscillatory responses.



FORM OF RESPONSE

- ★ $b^2 - 4mk > 0 \rightarrow$ two real poles
- ★ $b = 2m\omega_n \rightarrow$ repeated real pole
- ★ $b^2 - 4mk < 0 \rightarrow$ complex pole pair

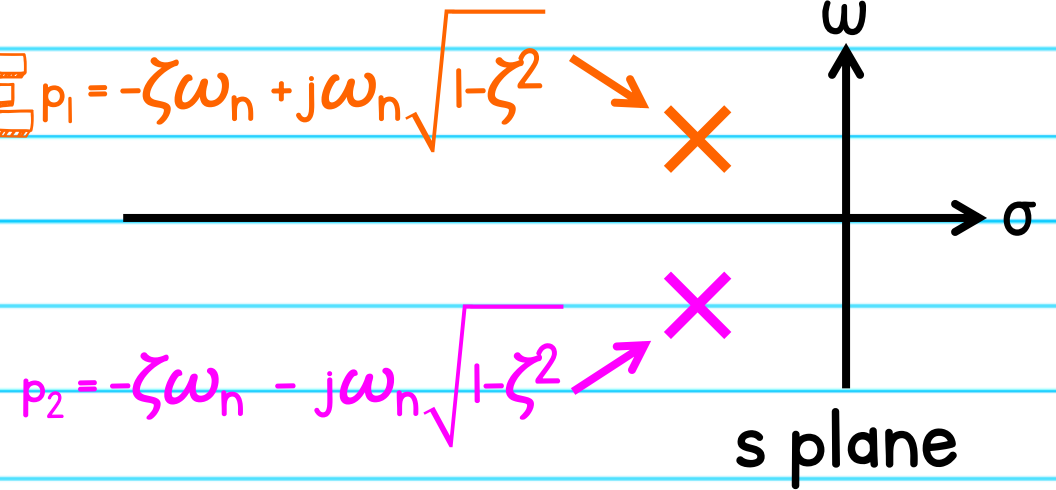


$$b^2 - 4mk < 0$$

$$x(t) = e^{-b/2m t} \cos\left(\frac{\sqrt{b^2 - 4mk}}{2m} t\right) u(t)$$

FORM OF RESPONSE

- ☆ $b^2 - 4mk > 0 \rightarrow$ two real poles
- ☆ $b = 2m\omega_n \rightarrow$ repeated real pole
- ☆ $b^2 - 4mk < 0 \rightarrow$ complex pole pair



$$b^2 - 4mk < 0$$

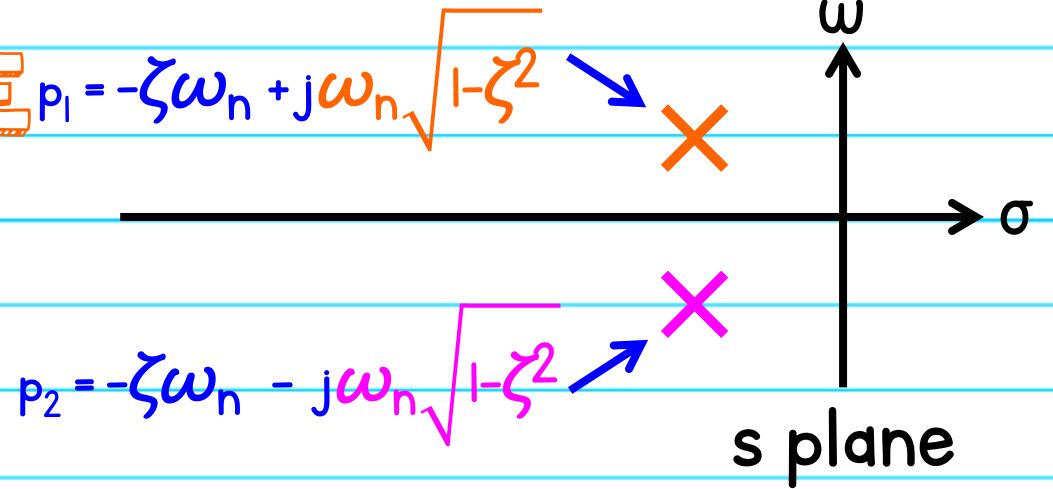
$$\omega_n = \sqrt{k/m}$$

$$\zeta = \frac{b}{2m\omega_n}$$

$$x(t) = e^{-\zeta\omega_n t} \cos(\omega_n\sqrt{1-\zeta^2}t)u(t)$$

FORM OF RESPONSE

- ★ $b^2 - 4mk > 0 \rightarrow$ two real poles
- ★ $b = 2m\omega_n \rightarrow$ repeated real pole
- ★ $b^2 - 4mk < 0 \rightarrow$ complex pole pair



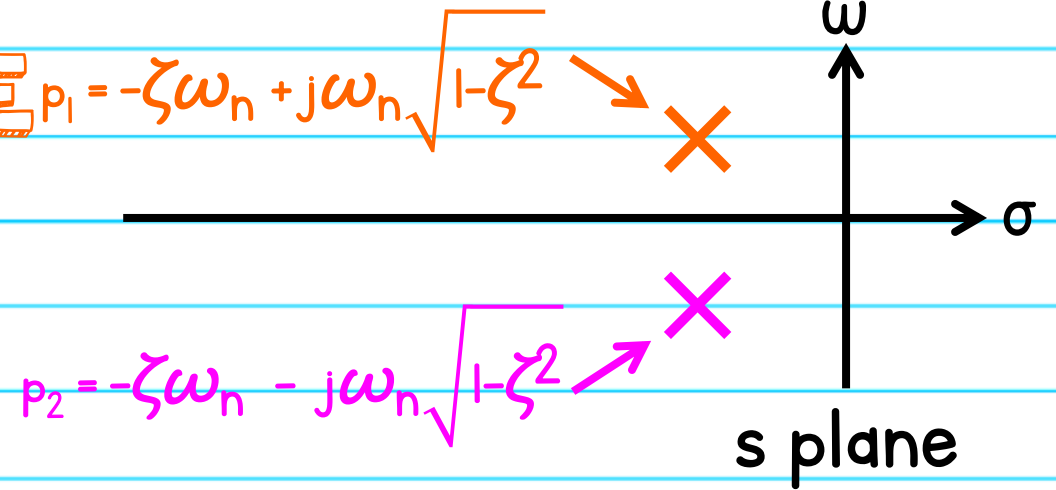
$$b^2 - 4mk < 0$$

The imaginary part of the pole gives the frequency of this oscillatory response...

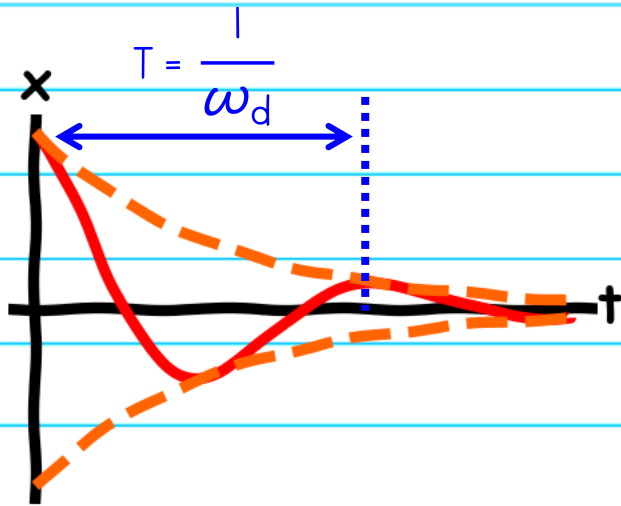
$$x(t) = e^{-\zeta\omega_n t} \cos(\omega_n\sqrt{1-\zeta^2}t)u(t)$$

FORM OF RESPONSE

- ★ $b^2 - 4mk > 0 \rightarrow$ two real poles
- ★ $b = 2m\omega_n \rightarrow$ repeated real pole
- ★ $b^2 - 4mk < 0 \rightarrow$ complex pole pair



$$b^2 - 4mk < 0$$



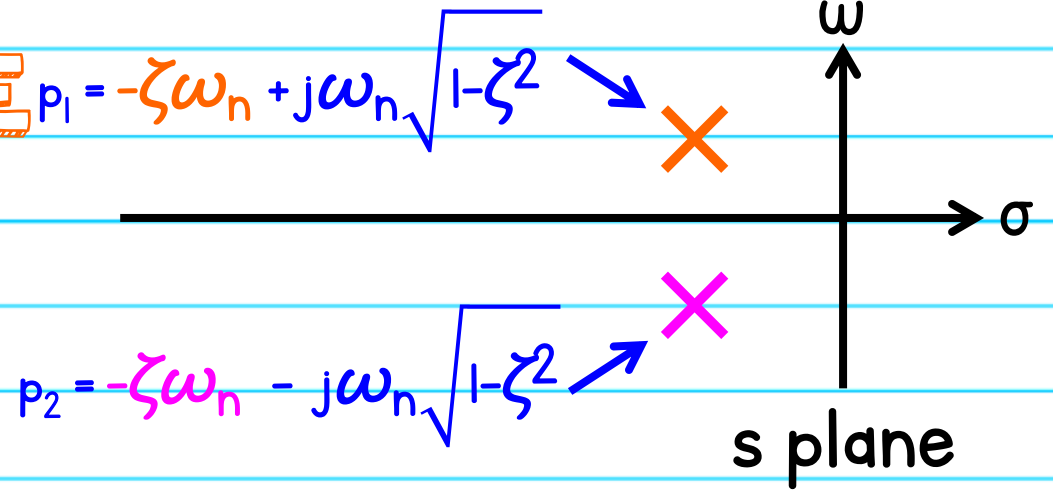
$$x(t) = e^{-\zeta\omega_n t} \cos(\omega_n \sqrt{1-\zeta^2} t) u(t)$$

damped frequency

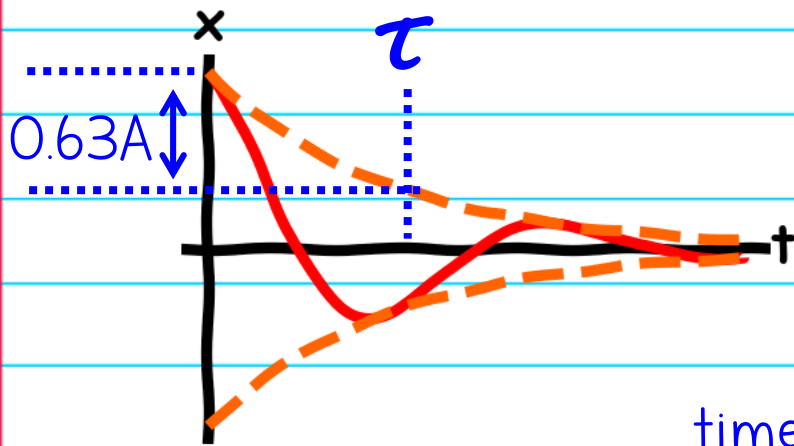
$$\omega_d = \omega_n \sqrt{1-\zeta^2}$$

FORM OF RESPONSE

- ★ $b^2 - 4mk > 0 \rightarrow$ two real poles
- ★ $b = 2m\omega_n \rightarrow$ repeated real pole
- ★ $b^2 - 4mk < 0 \rightarrow$ complex pole pair



$$b^2 - 4mk < 0$$



$$x(t) = e^{-\zeta\omega_n t} \cos(\omega_n \sqrt{1-\zeta^2} t) u(t)$$

time constant $\rightarrow$ (points to $\zeta\omega_n$)

damped frequency $\rightarrow$ (points to $\omega_n \sqrt{1-\zeta^2}$)

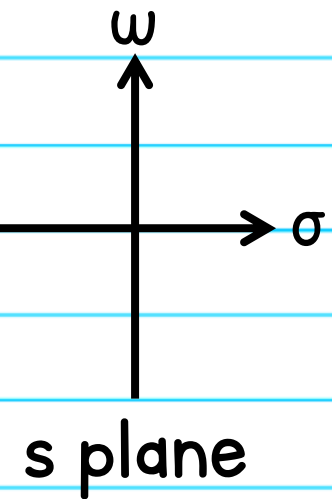
$$\therefore \tau = \frac{1}{\zeta\omega_n}$$

$$\omega_d = \omega_n \sqrt{1-\zeta^2}$$

...while the real part
of the pole gives the
time constant.

FORM OF RESPONSE

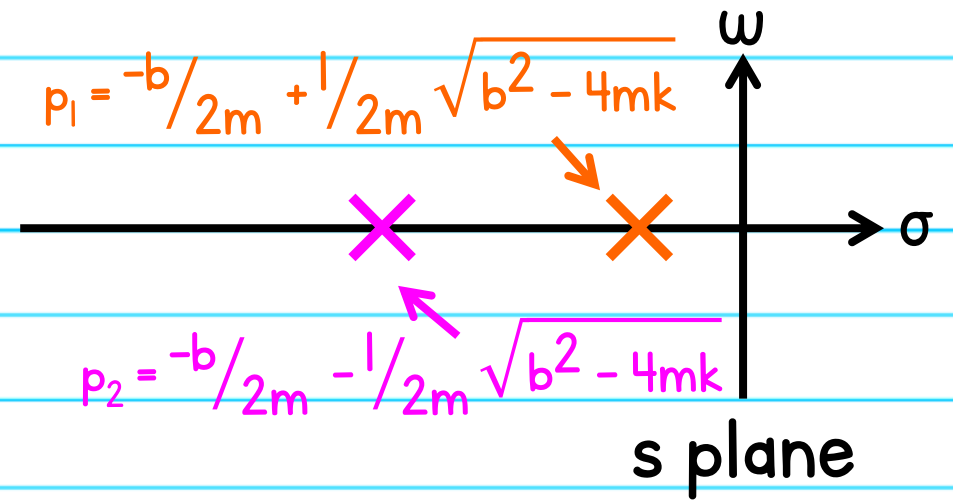
- ★ $b^2 - 4mk > 0 \rightarrow$ two real poles
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- ★ $b^2 - 4mk < 0 \rightarrow$ complex pole pair



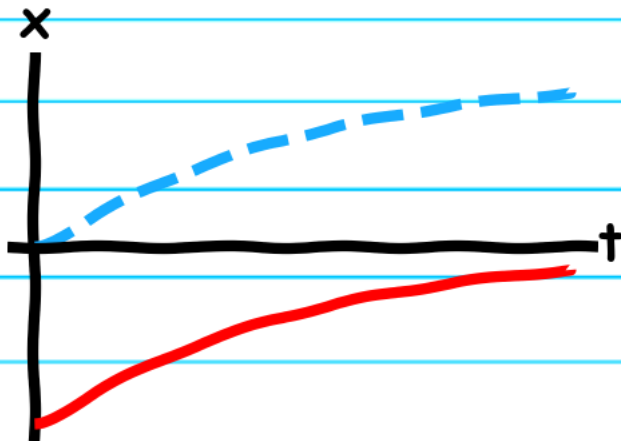
IN SUMMARY...

FORM OF RESPONSE

- ★ $b^2 - 4mk > 0 \rightarrow$ two real poles $\rightarrow$ overdamped
- ★ $b = 2m\omega_n \rightarrow$ repeated real pole
- ★ $b^2 - 4mk < 0 \rightarrow$ complex pole pair



OVERDAMPED



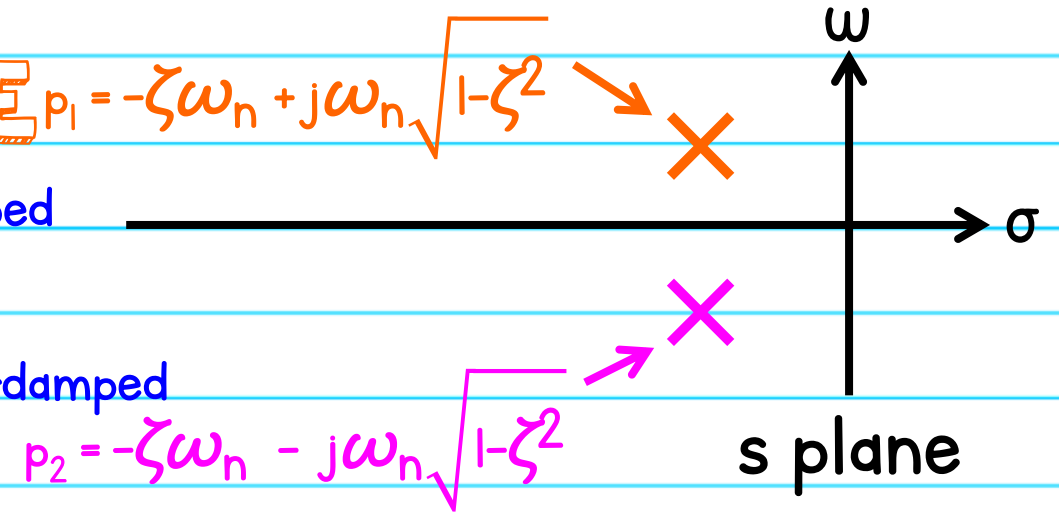
FORM OF RESPONSE

☆ $b^2 - 4mk > 0 \rightarrow$ two real poles $\rightarrow$ overdamped

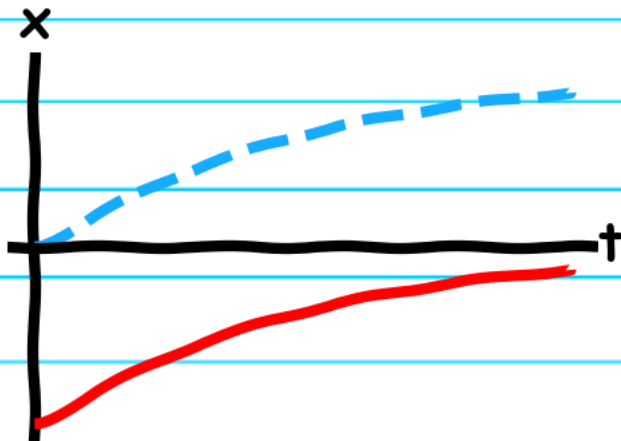
☆ $b = 2m\omega_n \rightarrow$ repeated real pole

☆ $b^2 - 4mk < 0 \rightarrow$ complex pole pair $\rightarrow$ underdamped

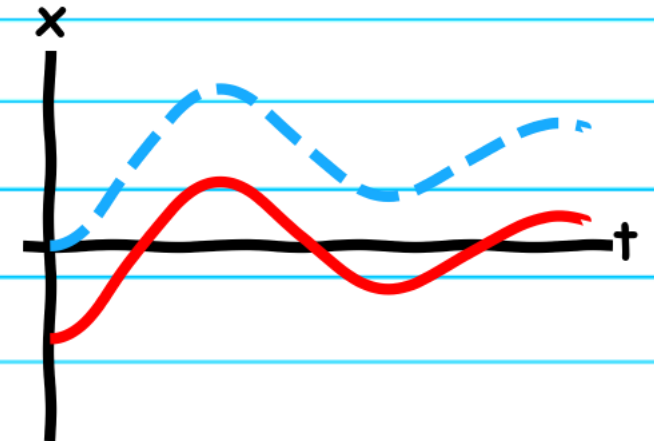
$$p_2 = -\zeta\omega_n - j\omega_n\sqrt{1-\zeta^2}$$



OVERDAMPED



UNDERDAMPED

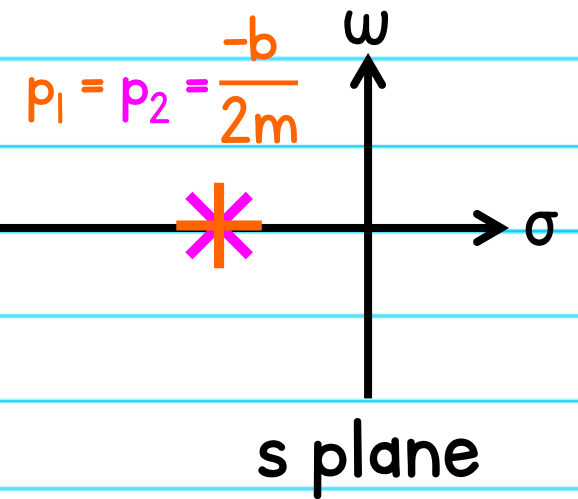


FORM OF RESPONSE

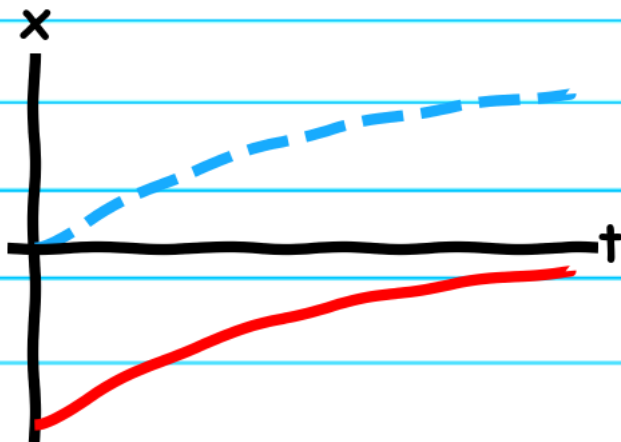
★ $b^2 - 4mk > 0 \rightarrow$ two real poles $\rightarrow$ overdamped

★ $b = 2m\omega_n \rightarrow$ repeated real pole $\rightarrow$ critically damped

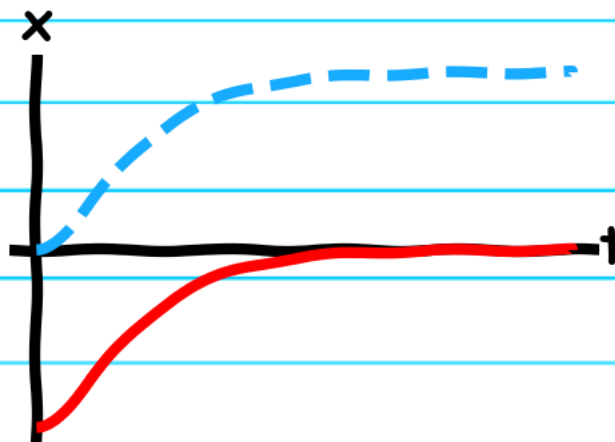
★ $b^2 - 4mk < 0 \rightarrow$ complex pole pair $\rightarrow$ underdamped



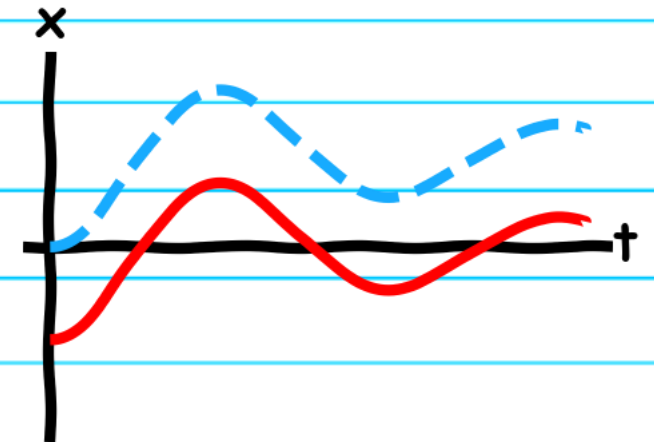
OVERDAMPED



CRITICALLY DAMPED



UNDERDAMPED



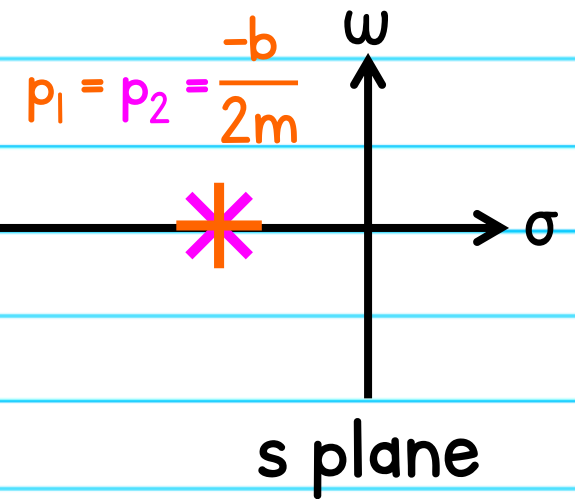
FORM OF RESPONSE

★ $b^2 - 4mk > 0 \rightarrow$ two real poles $\rightarrow$ overdamped

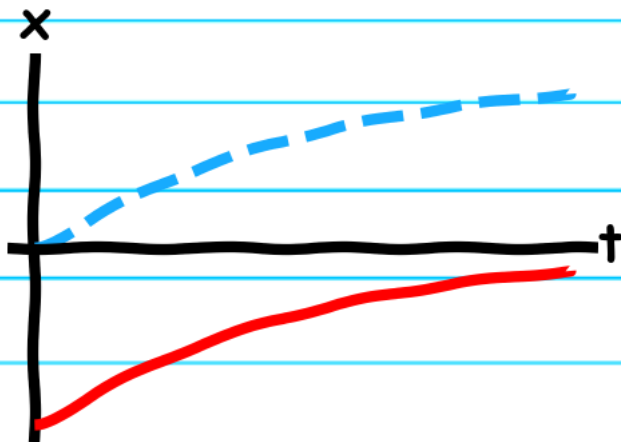
★ $b = 2m\omega_n \rightarrow$ repeated real pole $\rightarrow$ critically damped

★ $b^2 - 4mk < 0 \rightarrow$ complex pole pair $\rightarrow$ underdamped

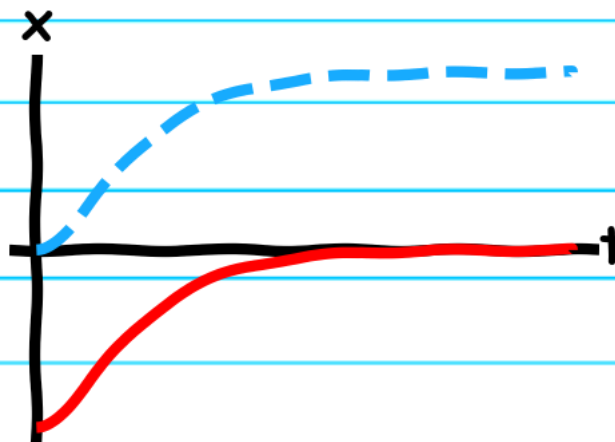
The critically damped response is the fastest possible response with no oscillation.



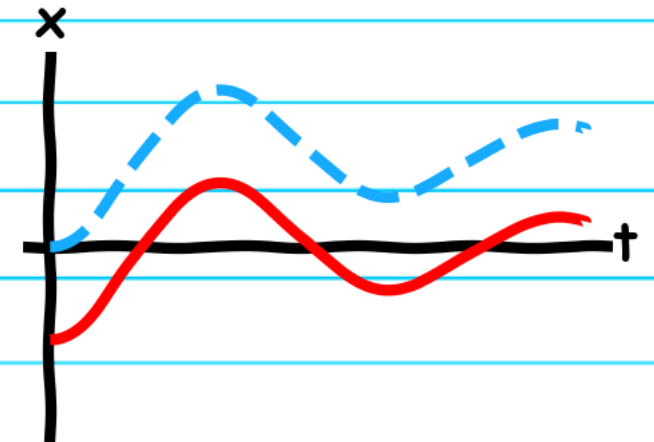
OVERDAMPED



CRITICALLY DAMPED



UNDERDAMPED



SECOND-ORDER SYSTEM

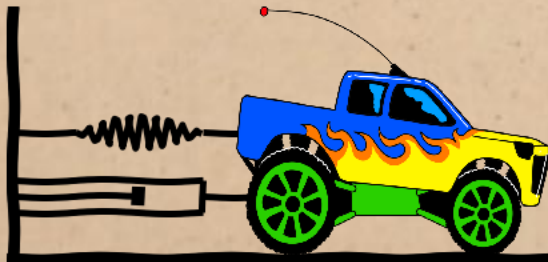
So the damping ratio indicates how close the actual damping of the system is to the value that results in a critically-damped response.

damping ratio

$$\zeta = \frac{b}{2m\omega_n}$$

$$A\omega_n^2$$

$$s^2 + 2\zeta\omega_n s + \omega_n^2$$



SECOND-ORDER SYSTEM

★ $b = 2m\omega_n$ → repeated real pole → critically damped

$\zeta = 1$ → critical damping

$\zeta < 1$ → underdamped

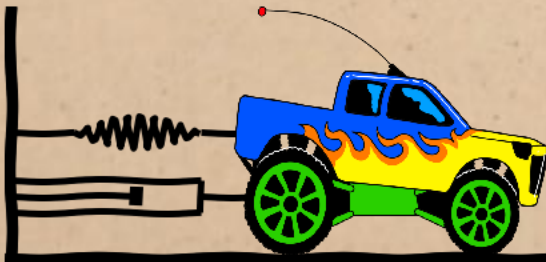
$\zeta > 1$ → overdamped

damping ratio

$$\zeta = \frac{b}{2m\omega_n}$$

$$A\omega_n^2$$

$$s^2 + 2\zeta\omega_n s + \omega_n^2$$



We can understand any complicated system, if we understand these simple modes.

